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Workpackage J
Innovative Tourism Statistics
Deliverable J3: Methodological framework report.

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1. Introduction

The use of new information sources (including big data sources) in official statistics opens up completely new possibilities of enriching and improving the system of tourism statistics. Thanks to external sources of information, the data provided by official statistics can be up to date and reflect the needs of users. Due to the differences between data from external sources and data from official statistics, a key element necessary to obtain valuable results is the development of methods for combining them.

This is a very crucial and at the same time difficult to implement element of workpackage J (WPJ) work, especially when it concerns the integration of data with big data characterized by a large amount of unstructured information. Therefore, in the short-term, WPJ aims to improve the completeness, timeliness and consistency of the survey frame of the tourism accommodation establishments.

This document describes methods used to implement the following tasks:

1. Combining data from different sources.
2. Flash estimates of occupancy of tourist accommodation establishments.

The results obtained using the implemented methods will significantly improve the quality of data on the capacity and occupancy of tourist accommodation establishments as well as trips of tourists and their incurred expenses. An important issue are also flash estimates of the occupancy of tourist accommodation establishments that will allow users to receive current information with high frequency at the lowest possible levels of aggregation.

This document consists of the following chapters:

Chapter 2 describes the methods used to combine data from various sources. It is divided into two subsections:

- subchapter 2.1 refers to the methods used to combine data from the frame for the survey of tourist accommodation establishments and data from portals offering accommodation booking. Implementation of the methods is illustrated on the examples of the Netherlands (2.1.1) and Hesse, DE (2.1.2).
- subchapter 2.2 refers to the methods used to develop estimates of trips and expenses of tourists. Implementation of the methods is illustrated on the example of Poland (2.2.1).

Chapter 3 contains a description of the method used to prepare flash estimates of tourist accommodated and the number of nights spent.

The report ends with recommendations related to the methods and tools used to combine data from multiple sources. Potential problems that may be encountered during the implementation of the presented methods were also described.

2. Combining data from different sources

The inventory of data sources described in the deliverable J2 – *Interim technical report showing the preliminary results and a general description of the methods used* revealed a large number of sources that are not yet being sufficiently used in tourism statistics. Sources with large information potential owned by external administrators, among others, data from smart city systems or web portals offering tourism-related services, play a special role. It is worth emphasizing that obtaining this type of data often requires many procedures and arrangements. Therefore, information that can be obtained by web scraping is crucial due to its public availability and wide reach.

2.1 Improvement of completeness and quality of frame in accommodation statistics

Scraped data and statistical survey data can be compared on the aggregated and individual level. One of the possible ways to use data from web scraping is to improve the frame of the survey of tourist accommodation establishments. This requires comparability on the individual data level. It can be achieved by the so-called data linkage.

Data linkage

Data linkage is a continuation of the work done as part of the WPJ package and its detailed description is presented in previous deliverables. The deliverable J1 presents sample portals and describes the variables that can be obtained from them whereas the deliverable J2 describes the data linking methods.

Data linkage (or data matching) enables to check if a given tourist accommodation establishment from the portals is also present in the survey frame. There are two main approaches to this task: deterministic (or rule-based) record linkage and probabilistic (or fuzzy) record linkage. In the case of the simplest deterministic matching, an identifier or a group of identifiers across databases are compared. Two records are linked when all of the identifiers agree. A single unique identifier, e.g. ID in Business Register, can serve as a matching variable. If it is not available then a group of identifiers can be used e.g. postal code and address. It may turn out that the combination of identifiers is not unique.

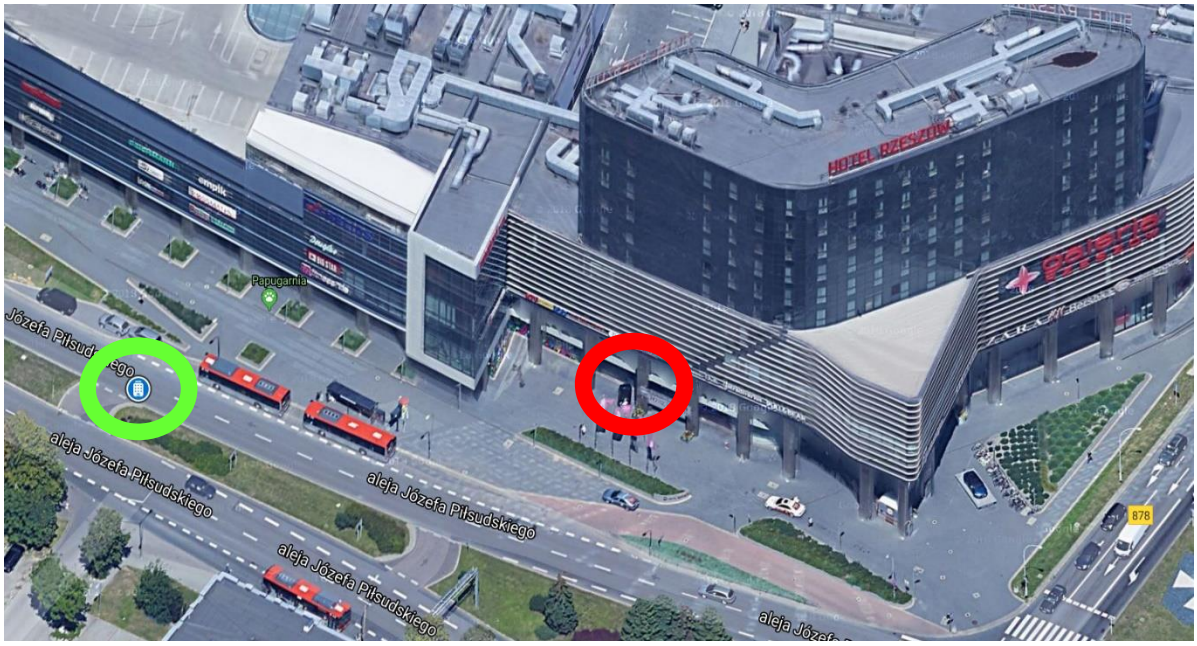
The scraped data contain several identifiers that can be used for the linkage:

- name of an accommodation establishment,
- address,
- postal code,
- longitude and latitude.

The linking process is however hampered by a number of difficulties. Postal codes and addresses from accommodation portals often contain typographical and data entry errors, such as transposed digits and misspellings. This situation may occur if the portal does not force a standard format when entering data (or uses a different scheme than the business register or addresses from the survey frame). Sometimes the address is hidden in the name of establishment while the field “address” is not filled in. On the other hand, geographical coordinates represented by pins on the map are sometimes absent or not placed precisely near the building entrance or in the centre of the building. Additionally, some hotels may choose their gate to the parking lot as their GPS-address. Figure 1 shows the difference

between the location of a hotel in hotels.com (green circle) and the location of the true entrance (red circle).

Figure 1. Discrepancy of establishment location



The distance between these two points is over 40 meters. As a result, there is no completely reliable identifier that can be used for linkage. Thus, the data linkage cannot be conducted without additional data pre-processing and additional tools.

Distance-based linkage

In the first step, addresses and postal codes need to be to the same format. Many rules standardizing text strings may be implemented with the so-called regular expressions. For more details see, e.g. Friedl J. (2006). Standardization is recommended for every type of data linking. Data standardization enables to analyse and use data in a consistent manner. Unfortunately, such standardization is difficult for addresses. Variety of names of streets and misspellings make it hard to describe all possible regular expressions to handle the problem.

This results in difficulty to perform deterministic linking based on addresses but is very helpful during next step of data linkage, i.e. deriving latitude and longitude. These coordinates can be obtained using any geocoding software. Some countries have the knowledge and tools needed to implement the geolocation process. For other countries WPJ developed a tool for geolocation of address of accommodation establishments from the survey frame and web scraping. The solution was prepared in JavaScript language and uses *HERE Maps API*. HERE Maps are precise and up to date, so they are often used for car navigation. The variety of available functionalities in the application, as well as extensive documentation and technical support, allowed to build a universal tool being a part of the prototype created by WPJ. A very important function used in the developed solution is an automatic parsing of address data to a common structure, which significantly reduces the time needed to prepare batch files for the used tool. The more work is done in address standardization, the better performance of *HERE Maps API* is achieved. In a case that a given address is not recognized at all by the geolocation tool, latitude and longitude can be taken from the portal. In order to allow analysis and selection of

the optimal tool each partner, while working on the geolocation process and preparing case studies, has chosen which tool will be best suitable for their situation.

If the geographical coordinates are available for the addresses from the survey as well as from the accommodation portal, then the distance-based approach can be applied. In this approach there is no need for the establishments from different data sources to have exactly the same coordinates. Nevertheless, the coordinates in all data sources must come from the reference coordinate system which uniquely define the positions of the points in space. It is common to use WGS 84 - which is also used in the Global Positioning System (GPS).

The distance-based approach can be applied in the following way:

1. Calculate the distance between all establishments from the survey frame and all establishments from scraped data. This can be done with, e.g. Haversine formulae or Vincenty's formulae[1].
2. For each establishment in scraped data find all establishments in the survey frame for which the distance between the establishment from scraped data and establishments from the survey frame does not exceed a threshold.
3. For each establishment in scraped data, match the closest one found in the second step.

In the second step, various thresholds can be chosen with respect to, e.g. accommodation type groups. For instance, hotels – in general – are larger than apartments or holiday homes. Moreover, hotels have sometimes more than one entrance which results in two different sets of coordinates.
Setting the threshold plays a crucial role in this method. Figure 2 presents an example of correct matching.

Figure 2. Establishment matched correctly despite different coordinates

[1] Haversine formula is based on spherical trigonometry whereas Vincenty's formulae is based on ellipsoidal trigonometry and requires to solve iteratively a system of (truncated) equations.

The same establishment had slightly different coordinates in the survey frame and in scraped data. Setting 20 m search radius enabled to find a match.

Evaluation of data linkage

Quality of matching can be checked by reviewing names and addresses of paired establishments and deriving the number of correctly and incorrectly matched and not-matched establishments.

After matching of accommodation establishments, four situations may happen:

1. Establishment was paired correctly even if the distance between the coordinates was not zero (TP).
2. Two establishments were incorrectly paired due to small distance between the coordinates (distance smaller than threshold) (FP).
3. Establishment was not paired correctly due to high distance between the coordinates (TN).
4. Two establishments were not paired incorrectly due to high distance between the coordinates (FN).

Results of matching can be summarized in the form of a confusion matrix., which is often used when evaluating the performance of a classification model (see Table 1). For more details see, e.g. Christen et al. (2007).

Table 1. Confusion matrix for data linkage results

		Actual	
	Total Population(all establishments in the portals)	Match (establishment is present in portals and survey frame)	Non-match (establishment is present in portals only)
Predicted	Match	true matches true positives (TP)	false matches false positives (FP)
	Non-match	false non-matches false negatives (FN)	true non-matches true negatives (TN)

Increasing the threshold (i.e. the distance between a pair of coordinates) generate a higher number of true positives and false positives and, at the same time, a lower number of true negatives and false negatives – it is a trade-off. To find an optimal threshold several indicators can be used.

Accuracy is a share of correct results in a total population:

$$\text{Accuracy (ACC)} = \frac{\text{TP} + \text{TN}}{\text{Total population}}$$

Accuracy can be used if the error of false match is as important as the error of false non-match. If the error of incorrect matching is more important, then F1 score should be used. It is defined as follows:

$$\text{F1} \quad \text{score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is a harmonic mean of precision and recall defined by:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Improvement of quality of frame for survey of tourist accommodation establishments

The database of establishments created by data linkage of the statistical frame and scraped data (hereafter referred to as *joint database*) is the final output of the methodology for data linkage. This database contains three subsets:

1. Establishments which occurred in the statistical frame and scraped data - fully described by both data sources.
2. Establishments which occurred in the statistical frame only - described by statistical data in a sufficient way.
3. Establishments which occurred in the scraped data only - relevant variables need to be derived.

There are several variables which may not be covered by the scraped data or are obtained by web scraping with uncertainty and possible misclassification, e.g.:

- Type of accommodation with respect to statistical classification.

  Information about the *accommodation type* is presented in the portals but this does not necessarily match the accommodation type in the statistical frame. A set of possible accommodation types in the portals usually overlaps only partially accommodation types used in official statistics. Moreover, accommodation establishment may be, e.g. labelled *hotel* in the portals even if it is not a hotel according to statistical definitions.
- Number of bed places.

  It is usually not possible to derive a total number of bed places directly from the portals by a simple query. The reason is threefold:
  - only some of the establishment's rooms are offered on the portals,
  - some of the rooms offered on the portals are already booked (not displayed when querying),
  - there is a limit on the number of adults in a query.
- Months in which establishment is operating.

If the above-mentioned information is not collected by the official statistics through a survey, then it must be estimated in a different way. Data on the type and capacity of a tourist accommodation establishment can be verified and cleared based on additional information available in offers on the portals through the model-based classification. Fortunately, scraped data contain auxiliary variables which can be utilized for modelling: type of accommodation with respect to portals classification, price, presence of an offer in a given month, location/region, star rating, guest review rating, total guest reviews. Often the portals contain information on the number of rooms or apartments, which can be used in the model or can be used to directly estimate the number of beds, assuming some average number of beds in a room by type of establishment. If information on the number of rooms or apartments is not available on the portals, then auxiliary information about bed places from the portals can be obtained by setting several queries to the portals with different number of adults starting from

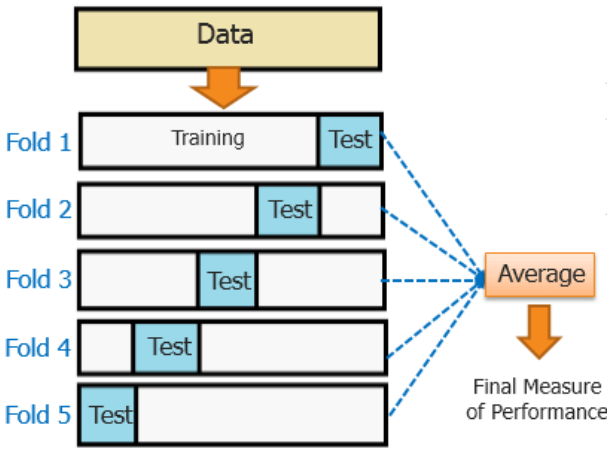
one and finishing on the upper limit. It enables to derive the maximum available number of bed places via the portals. Months can be derived from the presence of accommodation offer in a given month. If there is an offer, then the accommodation establishment operates in a given month. The reverse statement does not have to hold.

Preliminary analysis of scraped data revealed that on average hotels have higher prices, star rating, guest reviews rating and total number of reviews than other accommodation establishment types. Hotels also operate during the whole year more often than other establishments. Thus, all of these variables may be helpful to predict accommodation type for the establishments which are present only in the scraped data.

A subset of establishments in the *joint database*, common for scraped data and statistical data, contains information about the true accommodation type used in official statistics and months in which the establishment operates as well as variables obtained from the portals (listed above). This subset will be called the *training set*. One of the well-known classification method is a classification tree. Details of the method are provided in, e.g. Breiman et al. (1984). Thanks to the classification model, the type of accommodation used in official statistics can be assigned to all establishments that are only available in web scraped data. A similar procedure can be applied for assigning months in which the accommodation establishment is operating. The number of bed places can be estimated with a regression tree. Implementations of classification and regression trees are almost the same. Several implementations of machine learning (ML) methods in R can be found in Brownlee (2017).

Model selection in any ML method must be implemented carefully. For instance, if there are 2110 records in the *training set* and the size of the major class is 1252, then at the start the model prediction accuracy of 0.60 should be obtained classifying everything with the major class. Hence, the accuracy of the selected model must be higher than 0.60. A misclassification error rate is defined as the share of misclassified cases in the total number of cases. When a classification model is built, the relative error (rel error) and the cross-validation error (x-error) is generated. The rel error is a ratio of the misclassification error rate of the selected model to the misclassification error rate at the start. It always decreases when the complexity of the model increases. The x-error is produced by K-fold cross-validation. Figure 3 presents K-fold cross-validation procedure where K=5. The *training set* is divided into K subsets. One subset serves as a *test set* while the rest of the data serve as a *training set*. The relative error is computed for each fold using the *test set* and then mean (x-error) and standard deviation (x-std) is taken. The mean of x-error is called the cross-validation error.

Figure 3. K-fold cross-validation procedure (K=5)



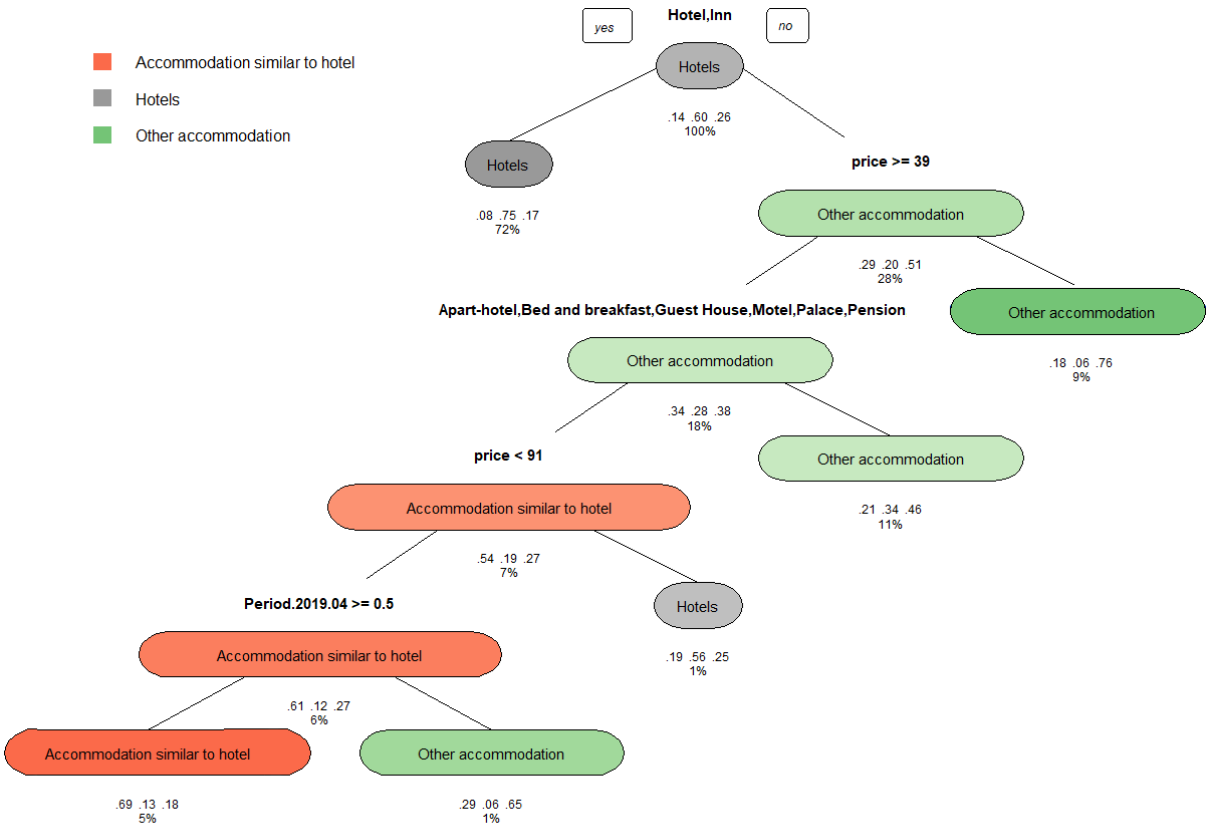
Source: https://www.edureka.co/blog/implementation-of-decision-tree/ [accessed: 29.01.2020]

In such an approach, the model is tested on the data that were not used to teach the model. Model selection based on x-error prevents overfitting. One of the thumb rule of model selection is: pick the most complex model such that its x-error does not exceeds minimal x-error + x-std.

Finally, after predictions are made, information about all required variables is available for all of the establishments in the *joint database*. Since geolocation data, the nomenclature of territorial units for statistics (NUTS), longitude and latitude coordinates are available, the data can be aggregated on any spatial level.

Preliminary results shows that the accuracy of accommodation type (hotel, accommodation similar to hotel, other accommodation) prediction amounted to 0.78 with relative error 0.549 and optimum complexity parameter 0.003. The next figure presents decision rules for assigning accommodation type group. A classification tree model was computed with higher complexity parameter (0.01) to provide an example (higher complexity parameter produced a smaller tree which is more visible).

Figure 4. Rpart.plot of classification model of the accommodation type



The relative error of predicting whether an establishment operates for the whole year or not was close to 1 which means that the classification model did not produce accurate results. Further studies will be focused on improving accuracy of the classification models.

2.1.1 Case study: the Netherlands

This case study shows the status of the web data extraction and processing associated to linking web scraped data to administrative and survey data for the Netherlands. Special attention is paid to keeping the outcomes of the project as close as possible to the statistical production of official statistics.

This section presents three issues:

- pre-processing of web scraped data from the hotels.com,
- results of address-driven (deterministic) linkage of web data to tourist accommodation establishments survey frame (yearly data),
- results of geo-driven (probabilistic) linkage of web data to tourist accommodation establishments data already matched to Register on Buildings and Addresses of the Netherlands (BAG).

Pre-processing of web data

Statistics Netherlands has been using two tools developed and deployed by Statistics Poland, namely a web scraper (*Hotel_com_v3*) and the geolocation tool since July and November 2019, respectively. Data gathered with these tools are meant to keep the tourist accommodation establishments survey frame up-to-date by:

a) supporting data collection,
b) foreseeing all data of (reverse) geo-data.

Pre-processing of web data is crucial to enable comparisons of accommodations within the database extracted from the *hotels.com* platform. Moreover, these data should be suited to compare them across other data sources and link them to enrich the survey frame. Pre-processing includes:

a) editing (standardization) of *hotel.com* data,
b) deduplication within hotels.com,
c) prioritization.

Main idea is to compare accommodations from *hotels.com* platform (and their features) to those from the admin survey data in a cost and time efficient way. In fact, the process of editing is a core process that enhances the comparability among accommodations.

Some specific features about *hotels.com* data

Before linking the data from *hotels.com* to the tourist accommodation establishments survey frame, statistical and geolocation features of the web scraped data need to be examined (see Table 2). It is worth to mention that the currently used version of the tool is *Hotel_com_v3*. This features depend mainly on the variable *location_id* and on the instances engineered by Statistics Poland, namely:

a) the number of guests in the reservations (1 or 2 adults and 0 children),
b) date of check-in.

Table 2 shows that the scraping approach executes three scenarios daily: *next day*, *next weekend* and *last Thursday of the month*. These options allow to monitor accommodations and their prices over time. Thus, the range of the hotel identifier is very wide and spans ID-values in the interval [75, 46293592]. The most forthcoming hotel names and their sub-classification of *hotels.com* can also be observed.

Additionally, the number of collected records (~85%) surpasses those of bed and breakfasts (B&B, ~4%). Further, about 1.3% of the records have an empty *accType*. It was observed that large resorts with hotel and holidays home options often leave this field empty. Besides, it may occur that reported accommodation types (*accType*) follow a rather arbitrary definition of the accommodation

by the owner/administrator of the accommodation. Sometimes the type of accommodation is filled-in based on the perception or marketing goals of the accommodation owner/administrator.

It is worth to mention that the cities of Amsterdam, Rotterdam and the Hague appear at the top of the list of *localities* advertising via the platform *hotels.com*. This situation is quite consistent with the levels of tourist accommodation establishments survey frame for the main touristic cities in the Netherlands.

The latitude and longitude values are quite complete *(~98.3%)*. About *(1.7%)* of the web scraped accommodations have no geolocations. Furthermore, the scraper seem to be collecting data outside the Dutch borders (longitude values < 3). It was also observed that the same number of records of the variable *postalCode* and variable *street* are missing address data. Given that the geolocations of these records are available, reverse geocoding[2] can be done on this subset of accommodations to obtain their corresponding *postalCode* and *street*.

When it comes to prices, it was noticed that the proportion of missing prices is relatively high (*17%*). Besides, the range of values suggest the presence of outliers. The outliers were checked and the number of accommodations involved amounted to 10. They are either luxury hotels or holiday homes offered at the beginning of summer (August) or before/during Christmas. There is even a B&B in this group. This can be a result of a typographical error. This column will need to be edited or these (*~600* or *~0.03%*) records left aside.

Moreover, the *total number of guest reviews* may be used as an auxiliary variable to determine whether an accommodation conducts its business, but also to prioritize the iterative control of accommodations during deduplication. Last, the *date* provides the period under study for which web scraped data of *hotel.com* were collected.

Table 2. Summary of web scraped records from *hotels.com*

Variable name	***Sort***	***Features***	***Needs Tokenization***
scrapingType	*Discrete*	last thursday of month:729736 next day :668295 next weekend :530093	No
hotelId	*Continuous*	Min. : 75 1st Qu.: 1146086 Median : 5344880 Mean : 9565551 3rd Qu.:15287372 Max. :46293592	Yes
hotelName	*Discrete*	Hotel The : 9321 Motel One : 5396 Motel Wa : 5078 Hotel In : 5013 M City Centre : 4940 B : 4537 (Other) :1893839	Yes
accType	*Discrete*	Hotel :1635753 Bed and breakfast: 81952 Apartment : 69919 Hostel : 49877 Guest House : 25451 (Other) : 39312 NA's : 25860	No

[2] Reverse geocoding the process of coding of a location (latitude, longitude) to a readable address or place name.

Table 2. Summary of web scraped records from *hotels.com* (cont.)

Variable name	***Sort***	***Features***	***Needs Tokenization***
locality	*Discrete*	Amsterdam : 480117 Rotterdam : 84257 The Hague : 60027 Maastricht: 57622 Eindhoven : 47495 Utrecht : 30334 (Other) :1168272	No
Lat, Lon	*Continuous*	Min. :50.47 Min. :2.895 1st Qu.:51.66 1st Qu.:4.688 Median :52.15 Median :4.898 Mean :52.04 Mean :5.098 3rd Qu.:52.37 3rd Qu.:5.701 Max. :53.59 Max. :7.575 NA's :25469 NA's :25469	Yes
postalCode	*Discrete*	undefined: 15771 1012LL : 11667 1011LZ : 9634 1171PK : 5927 5611CA : 5591 2132LZ : 5428 (Other) :1874106	No
street	*text*	Damrak NN : 9321 Stationsplein NN : 7789 Europaboulevard NN : 5396 Valkenburgerstraat NN : 5078 Lindengracht NN : 5013 (Other) :1879756 NA's : 15771	Yes
price	*Continuous*	Min. : 33.9 1st Qu.: 314.1 Median : 407.1 Mean : 525.7 3rd Qu.: 560.0 Max. :1782032.3 NA's :320419	No
guestReviewsTotal	*Continuous*	Min. : 0.0 1st Qu.: 23.0 Median : 76.0 Mean : 204.3 3rd Qu.: 230.0 Max. :3503.0	No
Date	*Ordinal*	Min. :2019-07-24 1st Qu.:2019-09-27 Median :2019-11-21 Mean :2019-11-13 3rd Qu.:2020-01-04 Max. :2020-02-03	No

Last but not the least, so far the scraper has collected 1.9 million records starting on July 24th 2019, i.e. a half-year data are available. There are a few gaps (missing data) caused by either improvement of the scraper or the collection strategy connected to the *destination-ID* which is required by the scraper to reach the *hotels.com* API. Statistics Netherlands use 141 *destinations-ID* based on cities and 12 *destinations-ID* on provinces. IT was noticed that when the scraper gathers data from the provinces

of Gelderland and Drenthe, some queries do not get response and the scraping process stops. This seems to happen at the beginning and/or end of each month. Table 3 shows that there are accommodations from Germany *(Aachen)* and Belgium in our records. These records have to be set aside as well.

Table 3. List of unique web scraped accommodations from *hotels.com*

Hotel Name	accType	locality	lat	lon	street	Guest Reviews Total
Hotel & Spa	Hotel	Nijmegen	NA	NA	Jonkerbos NN	60
Hotel De N	Hotel	Oss	NA	NA	Docfalaan NN	32
Hotel De W	Hotel	Oss	NA	NA	Oostwal NNN	39
B&B De H	B&B	Oeffelt	NA	NA	Heide NN	1
… … …

Thereafter, the set of variables that will deliver a unique number of accommodation establishments needs to be selected (see Table 4).

Table 4. Unique accommodations based on *hotelId*, *hotelName* and *accType*

hotelId	HotelName	accType
Min. : 75	City Hotel : 4	Hotel :2522
1st Qu.: 3954664	Ameland Rentals : 2	Apartment : 554
Median :12063802	Amsterdam Apartments : 2	Bed and breakfast: 442
Mean :15575212	ApartHotel N : 2	Guest House : 141
3rd Qu.:24055027	Aparthotel K : 2	Hostel : 104
Max. :46293592	Bed and Breakfast W : 2	(Other) : 267
	(Other) :4171	NA's : 155

It can be observed in Table 5 that accommodations may have more than 1 *hotelId* (54 cases were identified, there is even a case with 4 hotelIds).

Table 5. Accommodation with more than one *hotelId*

hotelId	HotelName	accType
10003291	City Hotel	Hotel
8514633	City Hotel	Hotel
37868525	City Hotel	Hotel
2179441	City Hotel	Hotel

It also seems that some accommodation establishments may be registered using two accommodation types (33 cases were identified). The example is presented in Table 6.

Table 6. Accommodation with two different *accType*

HotelName	accType
Parc B	Hotel
Parc B	Castle

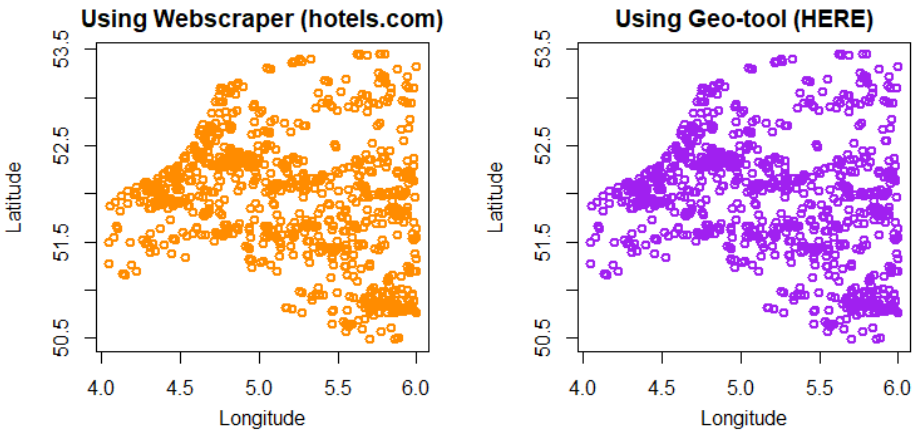
At first, 4009 unique accommodations were identified based on unique postal code and street and 4111 unique accommodations based on geolocation. However, part of these records has no postal address or the address is incomplete.

The remaining unique accommodations (including also the elimination of addresses in Germany or Belgium) amount to 1857, of which 197 accommodation are considered as new and will be added to the survey frame in 2020 once they have filled out their survey data. Part of these accommodations (the nonresponse part of the survey) needs to be checked by an expert to determine whether they are (or not) an accommodation.

Using geolocation tool on web scraped data from *hotels.com*

Next, it was checked whether the coordinates provided by the *hotels.com* platform are comparable to those of *HERE Maps API* obtained by the geolocation tool. Figure 6 shows both the latitude and longitude values from *hotels.com* and *HERE Maps API*, respectively.

Figure 6. Use of geolocation tool for data from *hotels.com*



Mapping latitude and longitude coordinates and differences

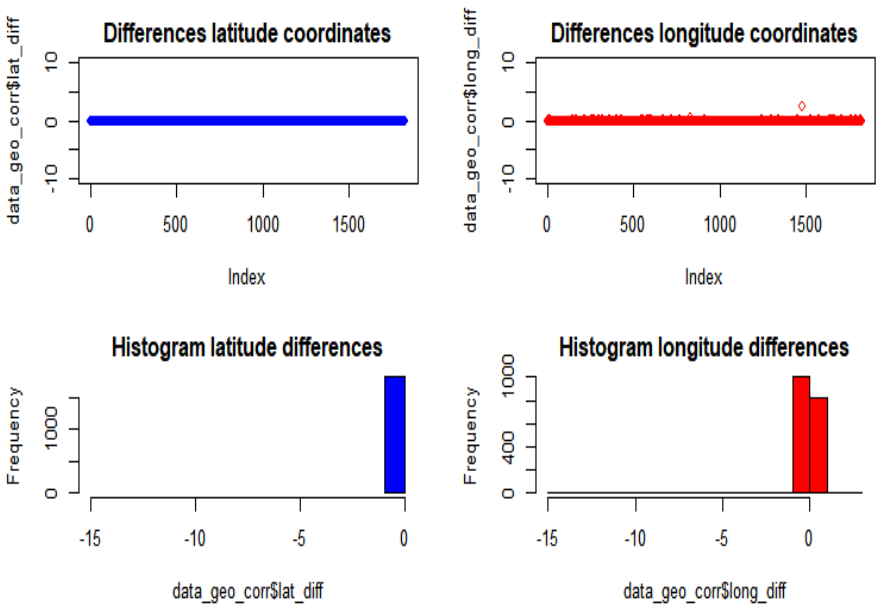
To determine the relative distance between the latitude- and longitude-coordinates obtained from hotels.com and those from the geo-tool, we have subtracted one of the other.

Table 7. Mapping latitude- and longitude-coordinates and differences

lat_diff	long_diff
Min. :0	Min. :-14.229838
1st Qu.:0	1st Qu.: -0.002950
Median :0	Median : -0.000021
Mean :0	Mean : -0.006299
3rd Qu.:0	3rd Qu.: 0.002527
Max. :0	Max. : 2.372678

In Figure 7 the latitude (in blue) and longitude (in red) coordinates gathered by the web scraper and t he geolocation tool were plotted.

Figure 7. Computed geo-locations using geolocation tool on data from *hotels.com*



The computed differences are based on the geolocation tool for data from *hotels.com*. The histogram shows that there are slight differences between the latitude and longitude values obtained by the geolocation tool and those of the *hotel.com* API. The latitude coordinates seem to be more sensitive than the longitude coordinates.

Checking geolocations and addresses from *hotels.com* and tourist accommodation establishments - BAG data

Figure 8 shows the accommodations collected via web scraping *(red)* and those of the tourist accommodation establishments survey frame already matched to the Register of Building and Addresses in the Netherlands *(orange)* in the island of Texel. A high density of accommodations in this touristic area is observed. When the island is zoomed-in (see Figure 9) and searched for a few addresses, then the BAG provides coordinates of the accommodations while the latitude and longitude in *hotels.com* refer often to a street/road. The yellow circles are the same accommodation, but with a slightly different pair (latitude, longitude) coordinates.

Figure 8. Study case = Texel island

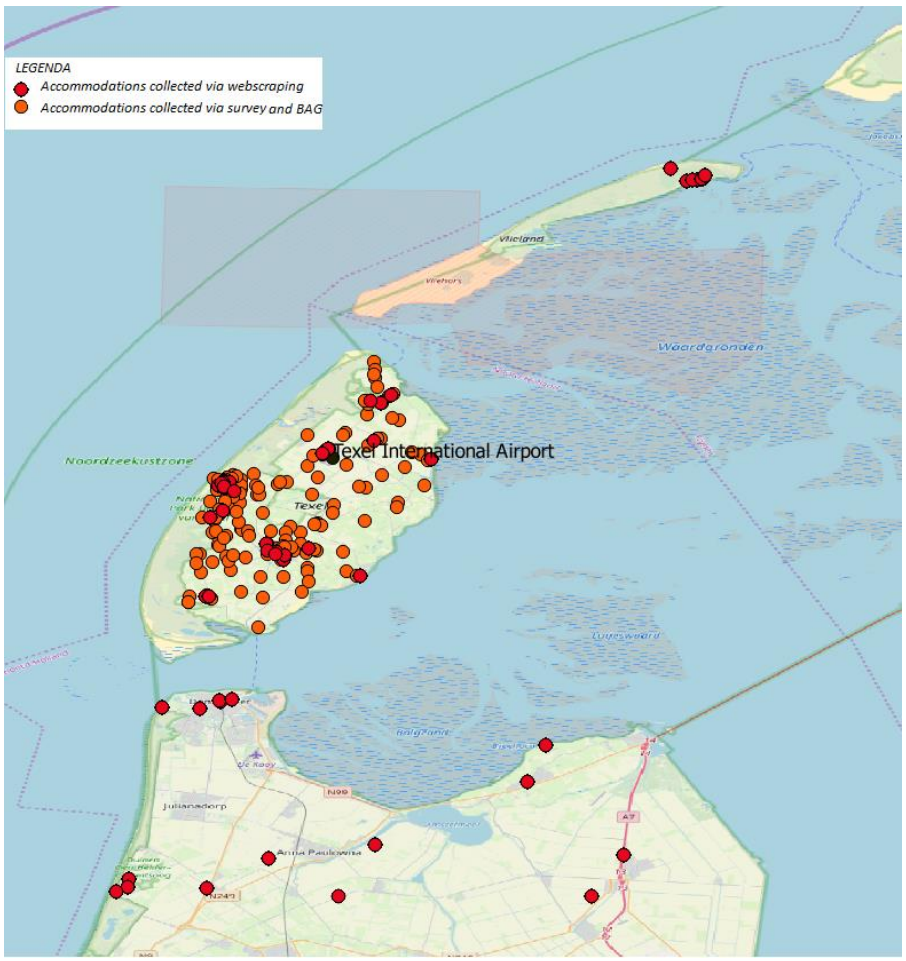


Figure 9. Examples of geolocation of accommodation in *hotels.com* and in tourist accommodation establishments survey frame

a) Grand Hotel Opduin

b) Boutique Hotel de Zwaluw

Last, it needs to be mentioned that Statistics Netherlands uses two methods to approach *hotels.com*. One via city halls (old method) and the other based on provinces (new method). In general, the city halls' approach gives in average 7000 records (with a maximum of almost 10000). The provinces' approach delivers in average 5000 records (with a maximum of 6000). The unique values are not much lower 1800 - 1900 (1858 accommodations in 2019). Although, this number runs almost parallel independent of the method, both methods are still used due to the fact that sometimes the web scraper does not get a response in some provinces. This occurs when scraping the provinces of Gelderland and Drenthe.

Results on *hotels.com* - tourist accommodation establishments

The (deterministic) linking between *hotels.com* and tourist accommodation establishments is straight forward and is based on the variables *postal code* and *house number*. The survey frame has two sets of addresses, one for the accommodation (location where the tourist spent the night) and another for the postal address. This is particularly useful for large hotel chains. In this case, the enquiry is often sent to the headquarters (postal address) of the hospitality businesses.
Both addresses were used to execute the matching. Thereafter, these data were concatenated and deduplicated based on the *AccomodationId*. *1540* matches were found.

Results on *hotels.com* - tourist accommodation establishments - BAG

The (deterministic) linking between *hotels.com*, the tourist accommodation establishments survey frame and the BAG is made based on *postal code* and *house number*. The matching deliver extra information about the building and the purpose of the dwelling (see Table 8).

Table 8. Matching web scraped, admin and survey data

Purpose of use of dwelling in Register	BAG Gebruiksdoel verblijfsobject	Frequency
Meeting	bijeenkomstfunctie	148
Health	gezondheidszorgfunctie	2
Industry	industriefunctie	7
Offices	kantoorfunctie	20
Accommodations	logiesfunctie	741
Education	onderwijsfunctie	3
Other uses	overige gebruiksfunctie	19
Sport	sportfunctie	4
Shop	winkelfunctie	6
Housing	woonfunctie	93
NA's	NA's	830

Next step is to try to match those accommodations that did not provide an *address* in *hotels.com*, but have only *GPS-coordinates*. That means using a probabilistic linking process. To this end, the use of Quantum GIS (QGIS, 2018) is considered.

Benchmark across NSIs

Table 9 gives an overview of the benchmarking of data provided by the WPJ countries after linking web scraping data to admin and survey data. Despite the differences in survey sizes, the number of new accommodations is relevant.

Table 9. Benchmarking of data provided by the WPJ countries after linking web scraping data to admin and survey data

Feature	**Slovakia_SK**	**Poland_PL**	**Hesse_DE**	**The Netherlands_NL** All accommodations	**TheNetherlands_NL** Hotels
Reference period	08.01.2020	11.2018-09.2019	15.10.2019	07-12.2019 01-02.2020	07-12.2019 01-02.2020
Survey Population	5930	10279	3483	8843	3808
Webscraping population	419	3956	731	1937	1857
1-1 matching	272	1080 (hotels)	415	1540	1212
New establishment found	92	142 (hotels)	125[3]	196	No data[4]

Coverage of Web, Admin and Survey data

Table 10 shows how the number of establishments in the survey depends on the season (or even on the month) the data are collected.

Table 10. Number of accommodations in survey frame reveals seasonality dependence (2019)

Total accommodation establishments	Total open accommodations	August	September	October	November
Nederland	number	8764	8748	7753	5904
Noord-Holland (PV)	number	1373	1368	1280	1105
Gelderland (PV)	number	1250	1249	1102	789
Noord-Brabant (PV)	number	959	962	861	659
Zeeland (PV)	number	876	875	782	440
Limburg (PV)	number	864	863	804	661
Friesland (PV)	number	832	825	685	494
Overijssel (PV)	number	745	743	603	431
Zuid-Holland (PV)	number	617	619	576	496
Drenthe (PV)	number	602	598	490	365
Utrecht (PV)	number	287	287	262	225
Groningen (PV)	number	256	257	218	176

[3] These negatives are not necessarily *new hotels* (false negatives) but, mostly true negatives, since the survey frame does not include small accommodation businesses by design.
[4] This value will be known once the survey 2020 is executed and checked by experts.

The distribution of the web scraped data and that of the tourist accommodation establishments inventory that has been linked to the Register of Business and Addresses (BAG) was shown (Figure 10 shows only the hotels from tourist accommodation establishments, 2019).

Figures 10 and 11 provide an overview of the geo-locations (sort of coverage) of the tourist accommodation establishments inventory (which has been linked to the Register of Business and Addresses, BAG) and the web scraped data of the tourist accommodation establishments advertising in hotels.com.

Figure 10. Distribution of tourist accommodation establishments survey frame in the Netherlands and the Wadden islands

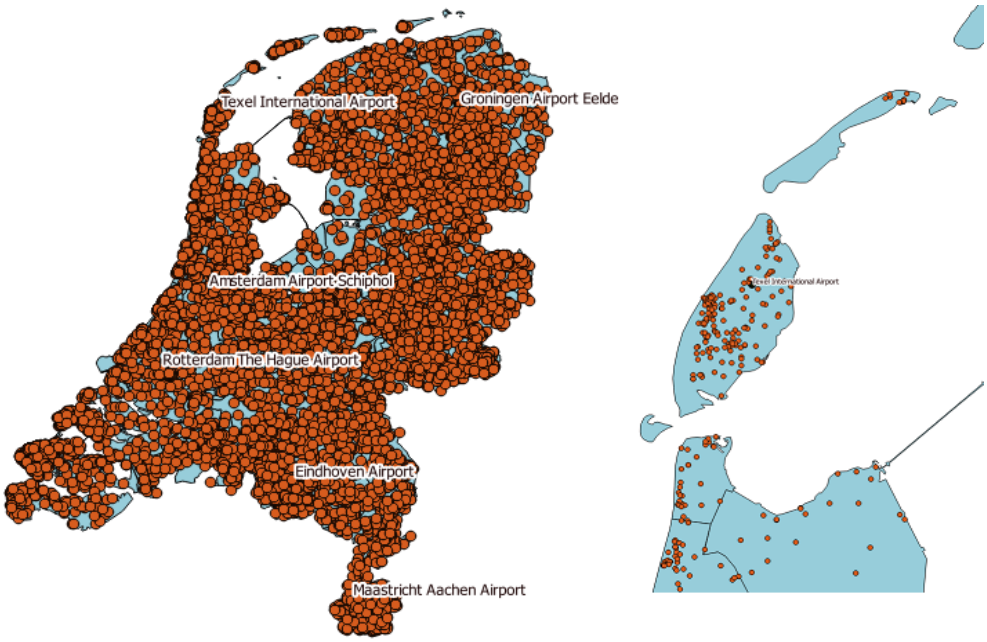
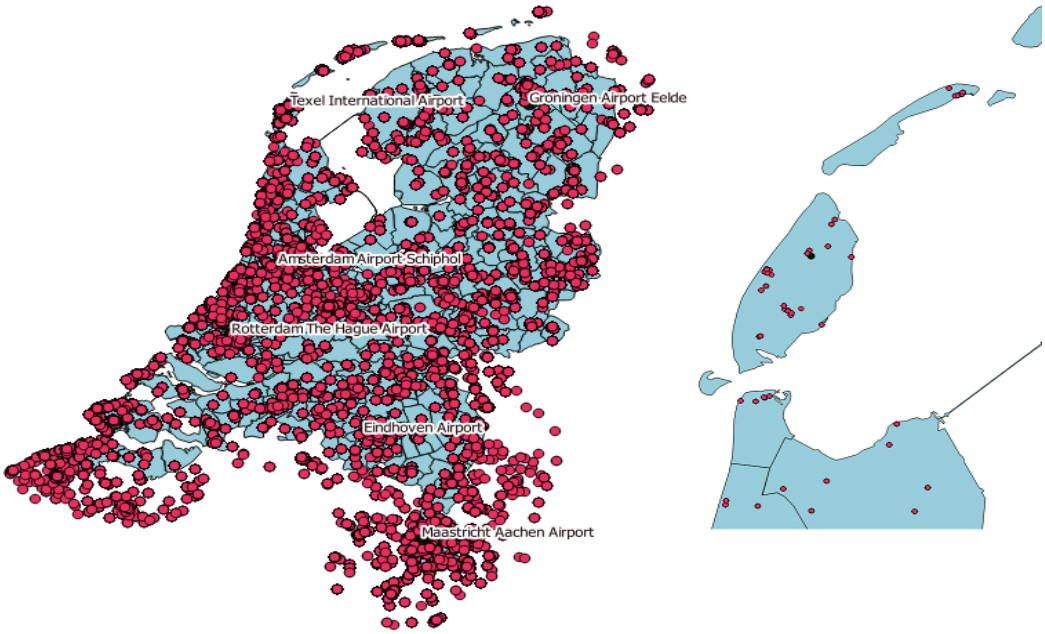


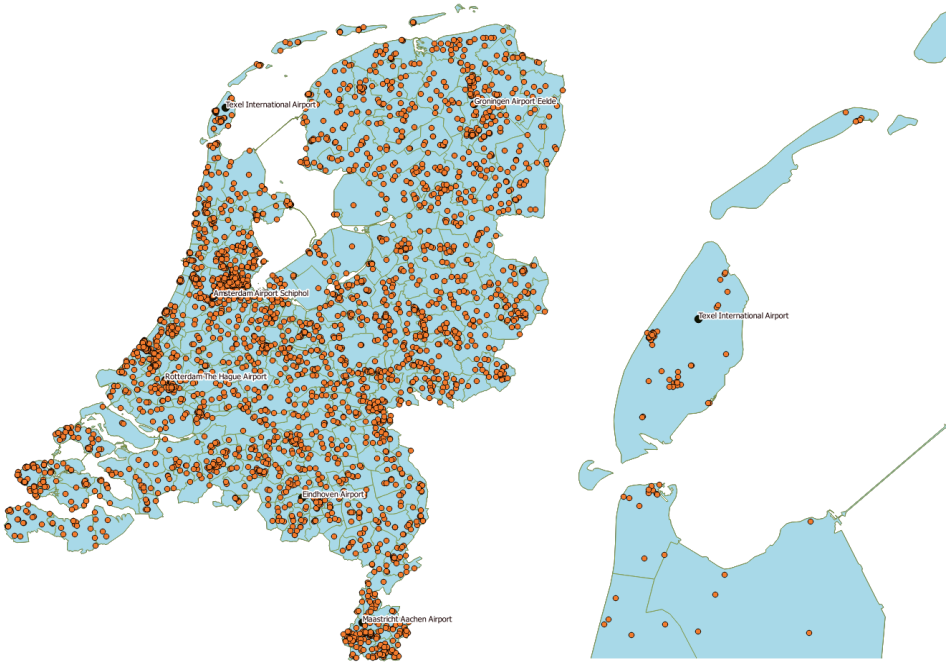
Figure 11. Distribution of hotels.com in the Netherlands and the Wadden islands (July 2019 – January 2020)



Notice that Figure 11 shows that the web scraper contains also foreign accommodations that have to be pre-filtered.

Figure 12 shows the distribution of accommodations from hotels.com and the hotels of the population of the survey of tourist accommodation establishments in the Netherlands and the Wadden islands. This figure shows the effect of matching web scraper data with Survey and Register data.

Figure 12. Distribution of hotels.com and Hotels population in the Netherlands and the Wadden islands (July 2019 - January 2020)



2.1.2 Case study: Hesse (Germany)

The following section describes some aspects of linking accommodations that stem from scraping accommodations from hotels.com portal with units from survey frame of tourist accommodation establishments.

In this case there is no common unique identifier available (e.g. a tax ID or a business register ID) but only names and addresses (note: which information is available for linking depends on the information given from the portal. For other portals, rather unique identifiers such as telephone numbers or an email address may be available).

In addition to address information (street name, house number, postal code, city name), often geographical coordinates of the accommodations' addresses are available. They can be derived from the source code of the webpage (e.g., from the marker on a map) or from the address using geocoding services (tools that are used in statistical offices anyway, or services like *OpenStreetMap* or several commercial map services).

Using coordinates either instead of or additional to address information has one big advantage. First, this is additional information that may serve as a linking variable. Second, typically, linking data requires a lot of pre-processing of the matching variables in order to make them as similar as possible (thus avoiding false negatives) without making them too similar (thus avoiding false positives). This is especially true for names and address information where different spellings or ways of writing down the same information may exist. In order to compare, e.g. street names, pre-processing of data has to be executed, taking into account abbreviations, additions to house numbers or number ranges for all

sources. Otherwise, exact comparisons will fail and result in too many establishments not linked (false negatives).

Using coordinates may facilitate matching in two ways: coordinates may serve as a way of standardizing addresses: tools to *reverse geocode* coordinates assign the address with a minimum distance to a given set of coordinates.[5] After *reverse geocoding* all pairs of coordinates from all data sources, the structure of the resulting addresses will be the same and comparison of the same addresses will not fail because of different spellings or ways of writing down parts of addresses. Second and more important, coordinates carry even more information than addresses and they can be used to calculate distances between a set of addresses. From two different addresses, it is not clear, whether the two establishments are close to another or not, e.g. *Luisenplatz 1* and *Rheinstrasse 41* in the city of Wiesbaden are very different addresses.

Figure 13. Two very different addresses with very low distance between coordinate pairs

When looking at the two maps in Figure 13, it can be observed that these two different addresses refer in fact to the same building (or at least are very close to each other). This may happen quite often at street corners or when addresses of the data sources have been generated automatically from coordinates.

Calculating and using distances as matching criteria become even more important since coordinates for the same building may differ between different sources due to technical details (e.g. algorithms, formulas and rounding) or different definitions of the position a given set of coordinates refers to (e.g. the main entrance, mid-point, a specific corner of an establishment).

Simple linkage of scraped accommodations from *hotels.com* with accommodations from the survey frame

The following section describe a very simple and deterministic linkage of accommodations from a booking portal with businesses from the tourism statistics survey frame in Hesse, Germany. Table 11 shows information that is available for linking data sources.

[5] See https://nominatim.org/release-docs/latest/api/Reverse/, e.g. https://nominatim.openstreetmap.org/reverse?lat=50.07808&lon=8.24122 which reverse geocodes the address of the statistical office of Hesse at the Rheinstraße.

Table 11. Information available for linking data sources

Scraped data from one hotel booking portal	Data from survey frame	Matched
Internal ID number (referring to unit or offer)	Internal ID number (anonymised)	NO
Accommodation name	Business name	YES
Accommodation type	Accommodations type	NO
Address information: street name and house number	Address information: street name and house number	YES
Address information: city name	Address information: city name	YES
Address information: postal code	Address information: postal code	YES
Address information: coordinate pairs	Address information: coordinate pairs	YES

Pre-processing of scraped data

First, data from the booking portal for a specific day or time period gets deduplicated by ID number which is used by the portal itself (alternatively, name and address information may be used), since each time, scraping is done for different point in the future (*next day*, *next weekend*, *last Thursday of the month*) in order to capture differences in prices. This means, that there are "duplicates" by design regarding unique units. For one specific date, there are 731 unique units available from the portal.[6] For these units, name, address (street name, house number, postal code and city name) as well as coordinates are available for linkage.[7] For 10 units, there is no street name available from the portal's site (and no coordinates, too).

In order to minimize efforts for standardizing address information, all units have been additionally geocoded in order to get identically structured address information.[8]

Regarding the accommodations' names, pre-processing consist of deleting or substituting special characters as well as umlauts or accents when it cannot be assumed that these characters are used in the same way in other data sources.

Table 12 shows the list of the most frequent tokens appearing more than five times in the names of accommodations from the portal. Of course, the token *hotel* is the most frequent one. Whether it is a clever idea to delete these frequent tokens from the name strings or not is not clear, since it may be useful to distinguish separate types of accommodations (and businesses) that might be distinguishable only by this token ("Hotel City Hall" vs. "Pension City Hall").

[6] Actually, much more *units* are scraped from the portal. It seems, that identical accommodations are either advertising within different destination cities, or the portal automatically shows accommodations from nearby cities when searching for accommodations within a specific destination.

[7] Coordinates stem from the portal site itself or from a geocoding services. The WPJ team from Poland has developed a tool to geocode addresses using the commercial *HERE Maps API*. Alternatively, free services like *OpenStreetMap* could be used, or tools that are available within statistical offices for these purposes (building registers). Note: no survey data has been geocoded using external tools.

[8] For test purposes, three different address-coordinate combinations have been used: information from the portal itself, geocoded addresses using the portal's coordinates using the tool developed by the WP J team from Poland, as well as the official building register for Hesse.

Table 12. List of frequent tokens in accommodation names (portal)

token	frequency	category / refers to
Hotel	362	type / name
Hof	32	name
Zum	24	stoppword
&	24	stoppword
am	23	stoppword
Frankfurt	23	location / name
Restaurant	22	type / name
Wiesbaden	19	location / name
Pension	18	type / name
Best	18	name / chain
Western	17	name / chain
Landhotel	16	type / name
Parkhotel	15	type / name
Kassel	15	location / name
Zur	15	stoppword
Bad	15	location / name
City	14	name
Airport	13	location / name
Landgasthof	12	type / name
Inn	11	name / chain
Haus	11	type / name
by	10	stoppword
Gästehaus	10	type / name
Villa	10	type / name
Darmstadt	9	location / name
Am	9	stoppword
Gasthof	9	type / name
Landhaus	8	type / name
ibis	8	name / chain
Dorint	7	name / chain
Krone	7	name
Mainz	7	location / name
Main	7	location / name
zum	7	stoppword
-	7	stoppword / special character
Romantik	6	name
ACHAT	6	name / chain
Apartments	6	type / name
Wetzlar	6	location / name
Fulda	6	location / name
Residenz	6	type / name
Altes	6	name
Plus	6	name / chain

Besides hints on the accommodation type, there often is a link to a location (which is not necessarily the accommodation's location or city name, e.g. hotel *Frankfurter Hof*, hotel *Fuldaer Hof*). In contrast to names coming from the business register, hints on the legal form are typically not part of an

accommodation's name from the portal. Finally, all parts of the name have been transformed into lower case letters.

Pre-processing of survey frame data

In principal, pre-processing consisted in exactly the same steps as for the scraped data. Regarding businesses' names, some business register specific parts of the name have been removed, because it can be assumed that these parts are never part of an accommodation's name when advertising on a booking site (e.g. names of the director of a business or the legal form of a business). For this purpose, names have also been split-up into separate tokens and occurrences of rather frequent and unique tokens have been counted. For a specific date (month), the survey frame consists of 3483 units.

Combining data from accommodation portal and survey frame

With only 731 units from the booking portal and 3483 units from the survey frame, there are 2546073 combinations possible.[9]

Combining data sources by address information

Table 13 shows the results after comparing all address components. Address information has to match exactly.

Table 13. Number of matches when matching address components

Matching variables	number of units from the portal	[in %]
Street name AND house number AND postal code AND city name	449	61.4
Street name AND house number AND (postal code OR city name)	0	0.0
Street name AND postal code AND city name	33	4.5
Street name AND (postal code OR city name)	0	0.0
Street name AND house number	13	1.8
Street name OR house number	191	26.0
No match at all	45	6.2
Total	731	100.0

For more than 61% of the 731 units advertising in the portal there is a match on all address components to businesses from the survey frame. In this case, there is no gain in relaxing the condition to either a matching city name or a matching postal code, since typically both fields are not filled independently. Another 33 or (4.5%) units match in street name and postal code (and city name). For them, without additional information (e.g. by looking at the name) it is not clear whether these are true matches or false positives. Since it is not very unlikely that there is more than one business located in one street –

[9] Even with moderate sizes like this, the complete Cartesian cross product of two files gets very large very fast. Therefore, it is desirable to reduce the set of combinations using a variable that is common to both data sources and that is not missing and is without any error. Such a *blocking variable* does not uniquely identify matching units but matches for all true matching units. Using such blocking variables (e.g. postal code or city name) reduces the number of pairs to compare from the full Cartesian product to a small fraction of it. In this example, no *blocking* is applied. Using a standardized city name as a blocking criterion would reduce the comparison space to only 23411 pairs (0.92%) instead of 2546073 without blocking.

it is not untypical that accommodations and other business cluster locally - so additional information definitely is needed and these cases are not considered matches at this point. For 45 units from the portal there is no match for any address information, for 204 units either street or house number match only.

Combining data sources by distance of addresses

For accommodations from the portal and all units from the survey frame coordinates are available, either directly from the portal or after geocoding addresses from the portal or the survey frame. With coordinates available, one could use coordinates in the sense of matching variables and compare their values. An even better way would be to use them to calculate distances between pairs of coordinates and to use this distance as a matching criterion.

Table 14 shows calculated distances between addresses for all 2.5 million pairs of addresses of both data sources and the number of accommodations from the portal by their *nearest neighbour*.

Table 14. Distance between all pairs and distance to the nearest neighbour to a portal's accommodation

distance (meters)	all pairs	in percent	portal units	in percent
d<=0	24	0.0	23	3.2
0<d<=1	381	0.0	379	51.9
1<d<=5	26	0.0	26	3.6
5<d<=10	12	0.0	11	1.5
10<d<=20	12	0.0	6	0.8
20<d<=30	30	0.0	9	1.2
30<d<=40	28	0.0	6	0.8
40<d<=50	34	0.0	8	1.1
50<d<=60	17	0.0	1	0.1
60<d<=70	29	0.0	0	0.0
70<d<=80	39	0.0	4	0.6
80<d<=90	43	0.0	3	0.4
90<d<=100	45	0.0	5	0.7
100<d<=150	224	0.0	8	1.1
150<d<=200	231	0.0	7	1.0
200<d<=300	500	0.0	5	0.7
300<d<=400	574	0.0	4	0.6
400<d<=500	555	0.0	2	0.3
500<d<=1000	2204	0.1	14	1.9
1000<d<=5000	16919	0.7	41	5.6
d>5000	2524146	99.1	169	23.1
Total	2546073	100.0	731	100.0

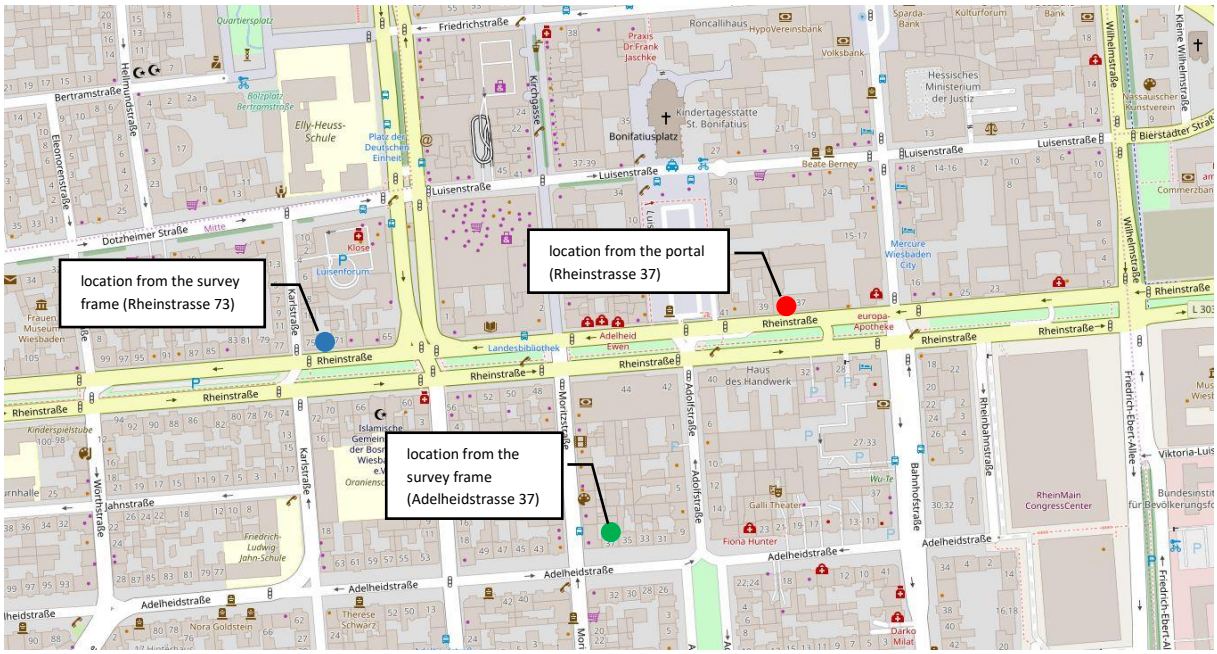
Given that addresses are complete and have no or only few errors (i.e. coordinates based on addresses are accurate), it can be argued, that the unit with the smallest distance corresponds to a true match (up to a specific threshold of the distance, say 70 meters).

For 379 units from the portal, there is a nearest neighbour from the survey frame within a distance up to 1 meter. For 469 of 731 (64.2%) accommodations from the portal, there is a nearest neighbour within 70 meters or less from the given set of coordinates.

If address data is incomplete or contains errors, there is a risk of assigning a wrong nearest neighbour. For example. when a house number was erroneously recorded as “#100” instead of “#10”, coordinates based on this erroneous address will differ accordingly. When calculating distances between pairs of addresses (i.e. coordinates) and choosing the closest pair as a match there may be a closer neighbour than the distance between house number #100 and #10.

When matching is based on distance only, the nearest neighbour may also be in a different street (see Figure 14).

Figure 14. False positive and false negative match when matching based on erroneous coordinates



Given there only is a typo in the address Rheinstrassse 37, 65185 Wiesbaden – marked by red dot on the figure above (instead of Rheinstrasse 73 – marked by blue dot) in one of two data sources, there is a nearer neighbour (based on raw distance between coordinates) in Adelheidstrasse – marked by green dot on the figure.

Combining data sources by matching coordinates

Coordinates could also be used as matching variables directly (instead of calculating distances): two units could be matched if their set of coordinates match. Whereas this still would give some flexibility regarding notation and spelling of addresses, there is the risk of many false positive as well as false negative matches.

For many reasons, coordinates may differ slightly between sources (due to different databases and their accuracy, different formulas and algorithms, different transformations, projections, re-projections and so on). For some applications, these differences in coordinates (and distances) may be very small and even neglectable or may not be of any practical importance. However, in many applications they are. For example, when comparing coordinates from different sources and using them for exact matching, the smallest differences matter, too.

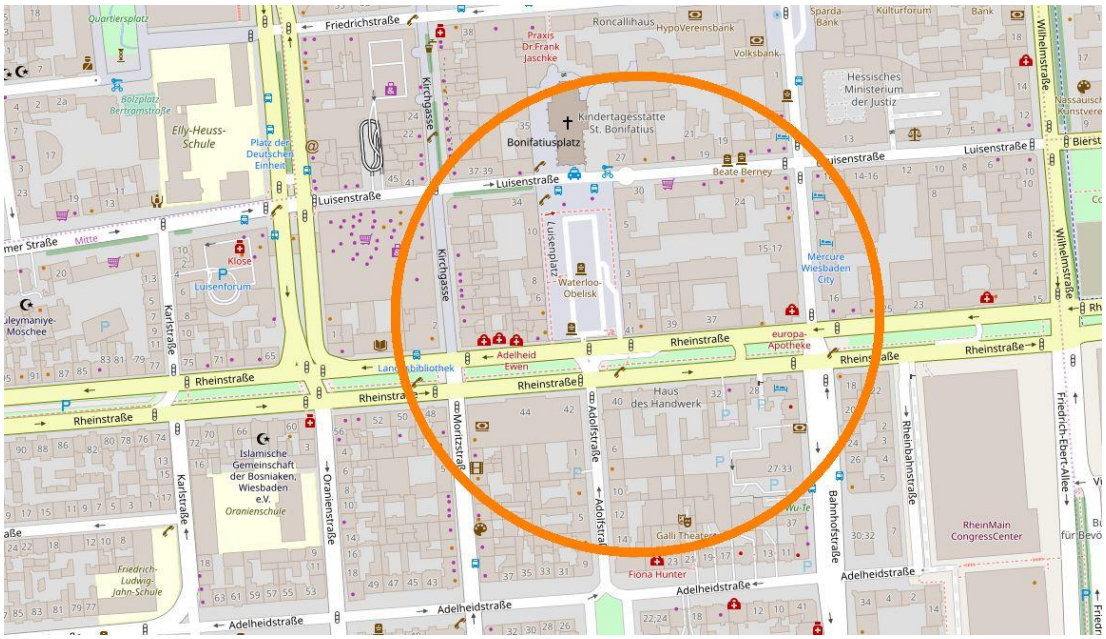
For example when taking the address *Rheinstrasse 37, 65185 Wiesbaden*, Google maps returns a set of coordinates of (50.07809, 8.2387913), whereas *OpenStreetMap* returns (50.07808, 8.24122). The calculated distance between these two sets of coordinates is about 174 meters (see Table 15).

Table 15. Distance in meters between coordinates of one address (*Rheinstrasse 37, 65185 Wiesbaden*) based on different geocoding services

	Openstreetmap	Google maps	HERE maps	BKG (building register)
	(50.07808, 8.24122)	(50.07809, 8.2387913)	(50.078121, 8.24303)	(50.0780108913468, 8.241073958631134)
Openstreetmap	0	173.85	129.63	12.98
Google maps	173.85	0	303.42	163.63
HERE maps	129.63	303.42	0	140.55
BKG	12.98	163.63	140.55	0

Even naive rounding should not be applied to coordinates in order to use them for exact comparisons. For example, rounding coordinates to the fourth digit after the comma (e.g. 50.078_, 8.241_) already covers a distance between two points of more than 100 meters. This may introduce additional possible matches (thus avoiding false negatives, but the decision will have to be made between several possible matches) Moreover, this naive rounding does not correspond to something like a *search area* with a specific radius/maximum distance around some set of coordinates. In fact, using some specific distance or circle around every pair of coordinates (see Figure 15) would be a very promising blocking criterion, since it could be more detailed or fine-grained than city name or postal code (thus being more efficient in terms of fewer observations to compare within each *block*). On the other hand, it could be less efficient than city name or postal code, since every pair of coordinates represents its own block (so there will be many blocks) and these blocks are not disjunct by design anymore but may overlap. In order not to introduce many false negatives, these circles must be large enough. Consequently, many units may be a member of more than one block.

Figure 15. Blocking by distance given a set of coordinates



So in general, coordinates are very useful for linking data sources but should not be used naively or for exact matching but for calculating distances between establishments. In order to avoid some problems when comparing coordinates or avoid distortion when calculating distances, it would be advantageous, to only use coordinates from the same source (meaning that addresses for one data source have to be re-geocoded even when coordinates would already be available from the site itself).

Combining data using name information

Typically, businesses' names are composed of several parts or tokens and some pre-processing of names is necessary to identify these tokens and to standardize them before comparing. The purpose of pre-processing is to make similar names more similar without losing to much discriminating power of name parts that are important to separate matches from non-matches. Often, businesses names carry tokens that have low discriminating power, such as very frequent legal forms. In many instances, it increases discriminating power, when these frequent tokens are ignored/deleted (but are kept for final manual decisions, when necessary)[10]. On the other hand, pre-processing has to take care of name tokens that are supposed to have very high discriminating power (because they are very infrequent) but are supposed to appear only in one side of the data sources and would disturb comparisons. For example, the name field from the survey frame not only often contains the legal form of the business, but also the name of the director, the name of the holding and so on. The name field from the accommodation portal on the other hand typically does not contain such information, as it is not necessary for the user of this service.

Another obstacle with names consisting of many tokens is that there often is no clear rule or consistency for the order in which these token appear in the name field. For example, *Hotel ABC* and *ABC Hotel* may both be valid names and notations of the same establishment in different data sources. But when looking for equality of names, order may matter.

Additionally, basic pre-processing takes care of special characters, umlauts, accents, standardized upper and lower case and the like. But still, there are many ways of writing down an accommodation's name when advertising on a portal or when legally registering a business. Consequently, exact comparison of names will fail in many instances. For example, among all 2.5 million possible combinations of accommodations from the portal and the survey frame, there are only 157 combinations where names agree exactly.[11] Therefore, similarity measures can be applied that overcome the necessity of exact agreement.

For example, *n*-gram based methods split up character strings into small pieces of *n* characters and count the number of agreeing n-grams. In the case of n=3 (trigrams), *hotel abc* becomes the set of trigrams *[__h, _ho, hot, ote, tel, el_ l__, __a, _ab, bc_ c__]* and for all combinations agreeing n-grams are counted[12].

[10] Such decisions of deleting tokens should be taken carefully. Even when only one side contains these tokens, comparisons may not be largely affected: e.g. similarities between names are decreasing for all comparisons, and this decrease may be even constant. Sometimes a standardized and short notation of these tokens is a better choice than deleting.

[11] Note: Among these 157 combinations with exactly agreeing names are already some false positives, since some accommodation names are quite frequent, *Hotel Krone*, *Hotel Adler*, *Hotel am Mark*, and exist in several cities. So there are even less true matches with exactly the same name.

[12] There are many similarity functions defined. Some of them incorporate phonetical aspects or focus on positional aspects; other functions additionally use weights to account for frequent tokens or n-grams and so on.

Similar to distance above, one could again argue, that the combination of units with the most similar name correspond to a match (again, using a threshold of some minimum similarity of names). Table 16 shows the number of portal units according name similarity to their "best match".

Table 16. Number of units from internet portal by similarity of names from survey frame

similarity (trigrams)	number of portal units	[%]
0.2<=x<0.3	1	0.1
0.3<=x<0.4	13	1.8
0.4<=x<0.5	23	3.2
0.5<=x<0.6	63	8.6
0.6<=x<0.7	139	19.0
0.7<=x<0.8	155	21.2
0.8<=x<0.9	148	20.3
>=0.9	189	25.9
Total	731	100.0

For 189 accommodations from the internet portal, there is a business from the survey frame with a similarity of 0.9 or higher regarding their names. For 117 units, their names match exactly. Note: this is a *best match* regarding names only, meaning *Hotel ABC* in city X and *Hotel ABC* in city Y are *best matches* in this sense, too.

Bringing it all together: name similarity and distances

It is clear that it is vital to use a combination of (small) distance between two units' addresses as well as (high) similarity of the units' names in order to decide whether this is a match or not. Table 17 shows the cross tabulation of name similarity and distance between addresses. Not all name – address-combinations are of interest. Of course, true matches are assumed to be found in the shaded area: with high name similarity and small distance between addresses. Note: there are many pairs that have a very similar name but also show large distances: Some names are quite frequent and can be found in many cities (*Hotel Post*, *Hotel Krone*, *Hotel City Hall*).Table 18 shows only the most interesting pairs: 1088 pairs of bets matches with either the highest name similarity or lowest distance between addresses.

In this case, trigrams (*n*=3) has been used and tokens have been padded with two whitespace characters at the beginning and end of each token.

Table 17. Number of units by name similarity and distance between addresses

	name similarity (trigrams)									
distance (meters)	<0.1	0.1<=x<0.2	0.2<=x<0.3	0.3<=x<0.4	0.4<=x<0.5	0.5<=x<0.6	0.6<=x<0.7	0.7<=x<0.8	0.8<=x<0.9	>=.9
0<x<=1m	1	3	2	8	12	28	51	74	91	135
1<x<=10m	0	0	0	1	1	2	2	6	16	10
10<x<=20m	3	0	1	2	1	0	2	0	0	3
20<x<=30m	5	3	7	5	4	2	1	0	1	2
30<x<=40m	14	2	3	0	6	0	0	0	1	2
40<x<=50m	12	6	2	3	3	3	0	1	1	3
50<x<=60m	2	2	5	1	5	1	1	0	0	0
60<x<=70m	8	6	5	5	2	2	0	1	0	0
70<x<=80m	14	4	5	6	6	0	0	1	1	2
80<x<=90m	12	8	6	6	5	3	1	1	0	1
90<x<=100m	10	6	9	10	5	2	0	1	1	1
100<x<=200m	128	69	74	99	48	22	6	5	1	3
200<x<=300m	133	90	101	105	43	20	3	2	3	0
300<x<=500m	343	141	215	232	129	48	15	3	1	2
500<x<=1000m	677	359	413	384	233	85	40	10	3	0
>1000	1254486	476100	394788	277253	106590	25274	5498	822	184	70

Table 18. Number of units by name similarity and distance between addresses (best matches only)

	name similarity (trigrams)									
distance (meters)	<0.1	0.1<=x<0.2	0.2<=x<0.3	0.3<=x<0.4	0.4<=x<0.5	0.5<=x<0.6	0.6<=x<0.7	0.7<=x<0.8	0.8<=x<0.9	>=.9
0<x<=1m	1	2	2	6	12	28	51	74	91	135
1<x<=10m	0	0	0	1	0	2	2	6	16	10
10<x<=20m	0	0	1	1	0	0	1	0	0	3
20<x<=30m	1	0	2	2	1	0	0	0	1	2
30<x<=40m	1	0	0	0	2	0	0	0	1	2
40<x<=50m	3	0	0	0	0	0	0	1	1	3
50<x<=60m	0	0	0	0	0	0	1	0	0	0
60<x<=70m	1	0	0	0	0	0	0	0	0	0
70<x<=80m	2	0	0	0	0	0	0	0	0	2
80<x<=90m	0	0	1	0	0	0	0	1	0	1
90<x<=100m	1	1	0	0	0	1	0	1	1	1
100<x<=200m	0	2	1	2	2	2	2	2	0	3
200<x<=300m	3	1	0	1	2	0	0	0	1	0
300<x<=500m	2	1	0	1	0	0	2	0	1	2
500<x<=1000m	3	5	0	1	3	2	2	2	3	0
>1000	102	38	32	30	26	59	104	91	44	25

When all of these 1088 *best match* pairs are inspected manually, 490 true matches between survey frame and portal remain, 67.0%.[13] 731-490=241 accommodations from the portal cannot be linked to the survey frame for different reasons:

- Linking information is erroneous or completely wrong (false negative): true matches differ significantly in matching variables. More and additional linking variables are needed.
- Units from the portal are not part of the survey frame. First, the unit may be missing erroneously from the frame (false negative). Keep in mind that in the case of tourism statistics many countries apply a threshold: in Germany only businesses offering more than 10 beds are included in the survey. So most of these unmatched units from the portal are small businesses that advertise in the portal but are correctly missing in the survey frame by design (true negative).

2.2 Combining data for trips and expenditures estimation in country breakdown

In almost every EU's country National Statistical Institutes conduct a household sample survey on participation of residents in trips. Due to the level of detail of a questionnaire or small subsample size for some destination countries, the research have large variability in expenditure as well as the problem of non-response related to expenditure by expenditure category with respect to some destination countries. If a country domain is not well represented then there may be no representative in a sample for some strata and the results, e.g. expenditures for business trips cannot be estimated.

Countries that are visited by a different route than air are usually very numerous in the sample survey. Most travel destinations that are rare in the sample survey are visited by air. Therefore, the emphasis was put on this mode of transport.

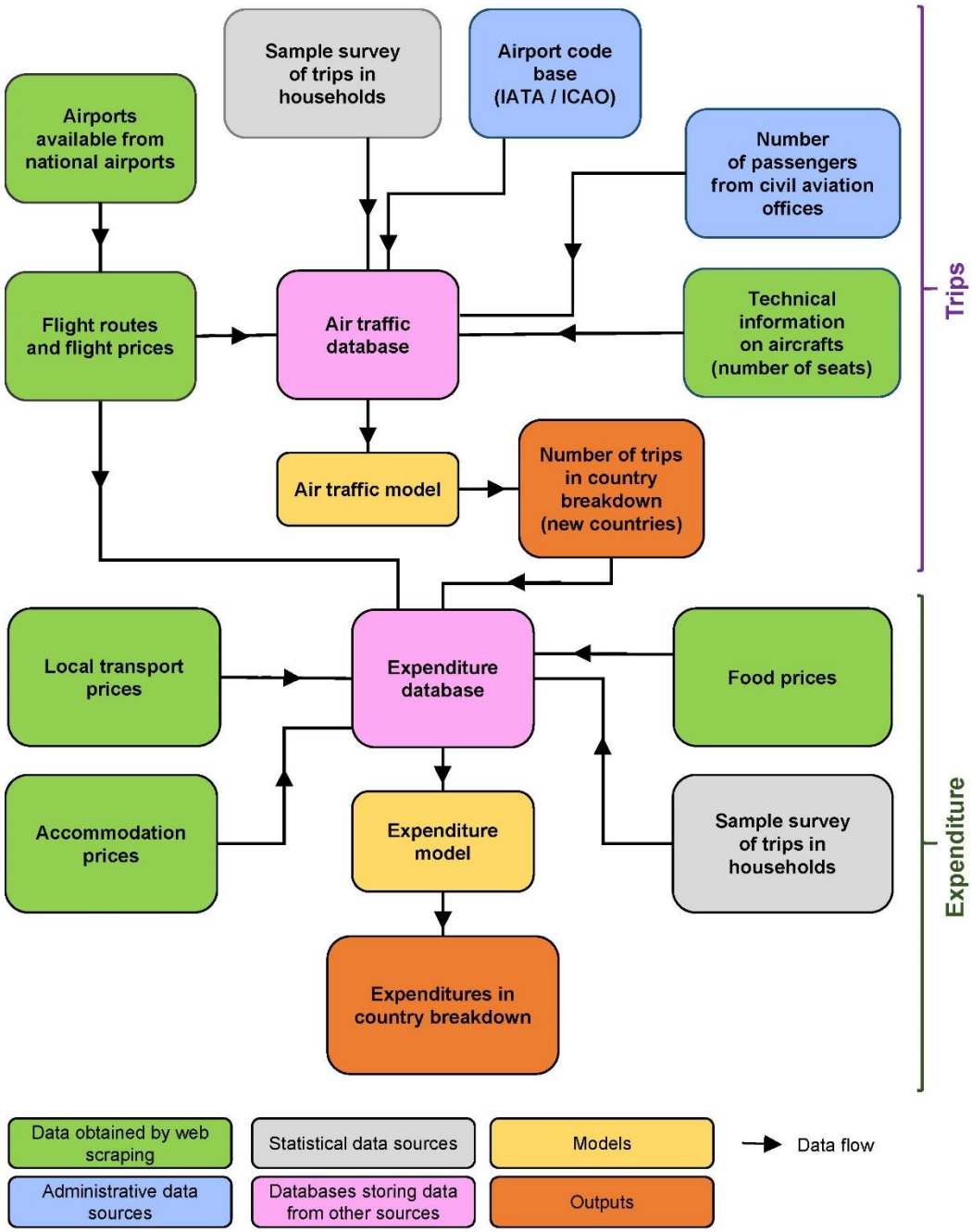
The methodology for estimating the number of trips in air traffic was developed in the Big Data I project. Within the framework of Big Data II project this methodology will be enhanced and linked to the methodology of expenditure estimation.

[13] These 490 linked accommodations could serve as a gold *standard* or *true ground* necessary for a more general approach, probabilistic record-linkage.

Data sources

The figure below shows data sources which can be combined.

Figure 16. Flowchart of trips and expenditure estimation



In the presented approach there are several data sources to combine to estimate trips and expenditures:

- destination airports available from the national airports,
- a list of airports all over the world covering airport name, city, country, International Air Transport Association (IATA), International Civil Aviation Organization (ICAO) and Federal Aviation Administration (FAA) codes,
- data on flight routes consisting of origin, hub and destination airports, IATA and ICAO codes, type of aircraft, date of arrival and departure,
- data on technical information on aircrafts about the type of aircraft, airline, ID, total number of seats and seats in each class,
- administrative data from Civil Aviation Offices (CAO) on the number of passengers from national airports in domestic and foreign traffic. Foreign traffic data covered only direct flights from national airports which gives much less countries than from the sample survey,
- crucial statistics of trips from the sample survey (share of residents in total international air traffic),
- prices and costs of flight tickets, accommodation, local transportation, restaurants and cafes, etc., obtained by web scraping of relevant portals,
- data on expenditure from the sample survey.

2.2.1 Case study: Poland

Methods of estimating trips

In the first step all possible routes to foreign destinations are collected. For this purpose online flight connection search engine is used. The key point is that this search engine, e.g. *skyscanner.com*, does not require the destination airport, only the airport of origin. In that way all possible destination airports available from national airports are gathered by web scraping. In this approach it is assumed that the vast majority of tourists use the national airports as a starting point.

These collected destination airports are used in another search engine – flight ticket search engine, e.g. kayak.com – as a search term. The search results provide information about hub airports, types of aircraft, airlines, departure and arrival data as well as prices of tickets. The flight ticket search engine is a component of data flow that crosses both trips and expenditure estimation.

Collecting possible routes in that way has an advantage over the method proposed in Big Data I. In the latter one, there was a need to manually exclude irrelevant routes that, with high probability, would not be used by passengers. For instance, all routes back to the country of origin, e.g. Poland-Germany-Poland were excluded. Also unlikely routes (e.g. Poland-France-Ukraine) were excluded when they were not time or cost efficient (since there are many direct flights from Poland to Ukraine. Comparing to the Big Data I solution, in that approach it seems to be easier to exclude irrelevant routes. The search provides sorting option: *cheapest*, *shortest* and *recommended*. The majority of passenger would probably pick one of the best offer in a given sorting. Thus, it is reasonable to discard all of the offers which are far from the best ones in a given category. The rule of discarding irrelevant offers may be of the form *keep top 20 offers* or *keep top 10% of offers*.

When the data on flight routes are collected, all airports are coded with IATA or ICAO codes. With the use of the code list the country where the given airport is located can be derived. It enables to calculate any statistics with respect to the country.

Data on flight routes also includes information about the type of an aircraft. Usually, the long-haul flights are operated by planes of high capacity, e.g. 300 seats. Information about the capacity of an aircraft is more valuable when it comes to passengers distribution among possible destination than the information about the flight solely. To assign the capacity of an aircraft, information about the airline which operates the flight is useful. The same aircraft model has various number of seats with respect to the airline. For instance, *Allegiant airlines* has Airbus A319 (319) with 156 seats while *Air China* has Airbus A319 (319) with 128 seats.

Finally, data on the number of passengers in a country breakdown is obtained from civil aviation offices. It may happen that there is no resident/non-resident breakdown in that data. Thus, relevant statistics may be taken from the sample survey of trips in households – precisely, the share of residents in the total air traffic. These data pertains only to the first country in the route. Therefore, it is used as a benchmark and it is not a part of the air traffic database. The share of residents using hub airports may be used to distinguish between passengers using a given airport as a final destination and the ones using a given airport as a hub airport if it is available from the sample survey of trips.

Since the search engine collects offers of flight tickets there are several offers for the same flight with various prices. Hence, in the air traffic database the flights are duplicated and they need to be removed. In that way, the distribution of flights is obtained. Then, if the number of seats is attached to the flight, the distribution of capacities is obtained. Definitely, using capacities of aircrafts would overestimate the number of passenger since the aircrafts are not usually fully loaded. Also, the share of used capacity varies among routes and hauls. To correct it, data on the number of passengers in air traffic in country breakdown obtained from civil aviation offices may be used. If data has no resident/non-resident breakdown, then the share of residents in the total traffic can be taken from the sample survey of trips to estimate the air traffic made by residents in a country breakdown. That data cover direct flights and only the first airport in the route. If the share of residents using hub airports is known, then the number of transit passengers and the number of passengers reaching their final destination can be derived. If it is not available then it must be assumed and calibrated latter according to outputs. In that way the number of passengers for each country provided by civil aviation offices is obtained including division into final destination airport and hub airports.

For instance, around 38 thousands of Polish residents flew to Austria in 2018, of which 23% continued their journey. Thus, 8.7 thousands (23% of 38 thousands) must be distributed among other countries. Air traffic database shows all routes starting with Poland-Austria. Proportionally to the capacities of aircrafts, 8.7 thousand is distributed among all flights departing from Poland to Austria. If there are two hub airports, the procedure is repeated. The final number of passengers for each destination country is calculated by summing up direct flights to a given country, the flights with one hub airport to a given country, the flights with two hub airports to a given country, etc.

The issue of concern is that the distribution of flights reflects preferences of passengers from all countries around the country of origin. It stems from the fact that the airlines operating in Europe meet the needs of many different routes. Therefore, using civil aviation offices data and search engines results which are also country-oriented, makes the results more specific to the country of origin.

In the last step, the total number of trips needs to be disaggregated into micro aggregates pertaining to categorical variables describing the trip, e.g. purpose of trip, type of accommodation, means of

transportation. For example, if trips of residents to a given country are described only by three variables. Information on the number of trips is available from historical data broken down by the previously mentioned three variables. This allows to specify the structure of trips to a given country as in the table below.

Table 19. Structure of trips of residents

Purpose	Accommodation establishment type	Trip with group tour operator or agent	Share [%]
Business	Hotel	No	32
Personal	Rented private flat	No	21
Personal	Lodging house	Yes	18
Personal	Hotel	Yes	15
Personal	Hotel	No	14

Assuming that the above structure of trips is derived from the past data of the sample survey the total number of trips obtained from big data sources may be distributed among these five groups by their shares. These micro aggregates are the basis of the expenditure estimation.

Methods of estimating expenditure

In the presented approach, there are two groups of data sources which are used to build the expenditure database: the sample survey of trips and data from various portals obtained through web scraping. The latter one can be further divided into data that should be collected frequently due to rapid changes, e.g. flight tickets prices, accommodation prices, and data that can be collected with low frequency, e.g. food and beverages, local transportation costs.

The data from the sample survey of trips should cover several consecutive years to include a wide range of trips with respect to their descriptions (e.g. destination country, purpose of trip, type of accommodation, means of transportation) and their expenditure.

Since some distant destination countries may not be well-represented in the expenditure database, auxiliary information is needed to improve the estimation of expenditure. Scraped data pertaining to flight prices of tickets and prices of accommodation contain plenty of offers including some breakdowns, e.g. accommodation establishment type. These data cannot be linked to the data from the sample survey directly. For instance, there is a trip to Brazil, means of transport: plane, accommodation establishment type: hotel. In the scraped data there are many offers of hotels in Brazil in a given period of time as well as many offers of flight tickets. Thus, the scraped data need to be aggregated: only some statistics should be derived from the data.

If the number of trips is already given, then only the average expenditure need to be estimated. The total expenditures will be obtained by multiplying the number of trips and the average expenditure by expenditure category.

In the first step the so-called *training set* must be prepared. It is a subset of the expenditure database of past data containing data for trips including descriptive variables such as destination country, purpose of trip, type of accommodation as well as the expenditure by expenditure category. There must be no missing values in the expenditure variables.

As mentioned previously, past data from the sample survey may not be sufficient. Hence, the scraped data covering several expenditure categories is used. These data need to be aggregated in the form of statistics that reflect the mean expenditures on accommodation or transportation for the country domain. According to the data analysis, it turned out that mean price is usually not the best option and some quartiles may serve better. At the beginning, several statistics pertaining to the given category of expenditures, period of time (e.g. quarter) and the country domain may be used, (e.g. the first quartile, median and mean). All of these statistics are attached to every trip in the expenditure database in the country domain and the period of time breakdown.

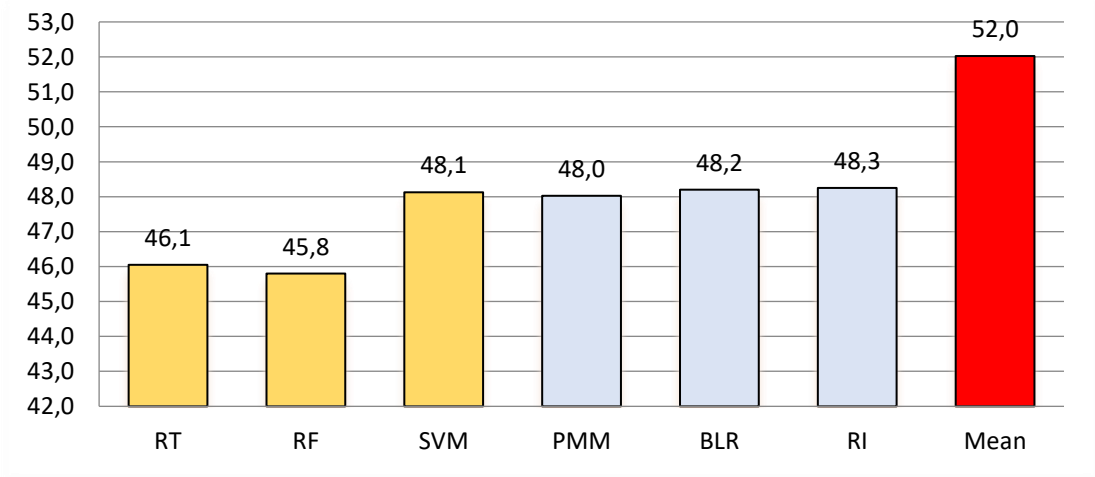
When the *training set* is prepared a method is run to predict expenditures. Among others, the following methods may be used:

- Regression Trees (RT) Breiman, et al. (1984),
- Support Vector Machine (SVM) Cristianini, et al. (2000),
- Random Forest (RF), Ho, (1998)
- Predictive Mean Matching (PMM), Rubin, (1986).,
- Bayesian Linear Regression (BLR), Box, et al. (1973),
- Random Indicator for Nonignorable Data (RI), Jolani, (2012).

These methods are available in *rpart, e1071, randomForest*, *mice*, and *caret* packages in R. Comparison and selection of the prediction model may be based on Root of Mean Squared Error (RMSE). It is possible to predict the expenditures for the new dataset, i.e. *test set*, with the use of the above mentioned model. This set is the set of micro aggregates which includes all auxiliary variables used in building prediction models. An example of the implementation of the expenditure estimation procedure will be included in the deliverable J5 - *Final Report containing final results and a full description of the methodology used* in the form of R script.

Aforementioned methods were tested on individual data for a prediction of the expenditures for accommodation. Imputation with mean served as a benchmark. Preliminary results based on RMSE showed that the best accuracy among the tested method was achieved by RT and RF as shown in the Figure 17.

Figure 17. RMSE of prediction of the expenditure for accommodation [in EUR]



Further studies will be focused on the estimation of other categories of the expenditures (i.e. restaurants and cafes, transportation and other services).

3. Flash estimates of occupancy of tourist accommodation establishments

According to the Regulation (EU) No 692/2011 of the European Parliament and of the Council of 6 July 2011 concerning European statistics on tourism as well as the Commission Delegated Regulation (EU) 2019/1681 of 1 August 2019 amending Regulation (EU) No 692/2011 of the European Parliament and of the Council concerning European statistics on tourism, European Union countries should submit to Eurostat data on accommodation statistics in the following scope:

1. For "hotels and similar accommodation" and for "holidays and other short-stay accommodation", the scope of observation should at least include all tourist accommodation establishments having 10 or more bed places.
2. For "camping grounds, recreational vehicle parks and trailer parks", the scope of observation should at least include all tourist accommodation establishments having 10 or more places.
3. Member States accounting for less than 1% of the total annual number of nights spent at tourist accommodation establishments in the European Union may further reduce the scope of observation, to at least include all tourist accommodation establishments having 20 or more bed places (20 or more places).

The scope of information about tourist accommodation establishments provided to Eurostat can therefore be and in fact is different between countries. In many countries establishments having less than ten bed places are not covered by the surveys, additionally different thresholds are used in specific countries to collect information on establishments. The frequency of data collection is not very diverse and usually data are collected on a monthly basis (in the case of project partners, all countries collect information on a monthly basis), while differences often occur in the observed population of establishments as well as methods of data collection.

This is an important issue when developing flash monthly estimates concerning capacity of tourist accommodation establishments as specific countries has a different scope of information for use based on surveys. Table 20 shows the scope and frequency of survey on tourist accommodation establishments in partner countries

Table 20. Data collection on accommodation establishments in partner countries

Country	Surveys on accommodation establishments	
	Monthly	Yearly
BG	Establishments with 10 or more bed places	
DE	Establishments with 10 or more bed places	
EL	Without threshold	
IT	Without threshold	Rural accommodation

Table 20. Data collection on accommodation establishments in partner countries (cont.)

Country	Surveys on accommodation establishments	
	Monthly	**Yearly**
NL	Sample of establishments is monthly drawn from yearly survey, which is based on opening periods and bounded to capacity restriction: 55.1 – Hotels, hostels and B&Bs with more than 5 bed places 55.2 – Rental houses/apartments/youth hostels with more than 10 bed places 55.3 – Camping sites with more than 4 pitches	Without threshold
PL	Establishments with 10 or more bed places	Establishments with 9 or less bed places
PT	Hotel establishments (hotels, apartment hotels, tourist apartments, tourist villages, pousadas and quintas da Madeira) – without threshold	
	Local accommodation with 10 or more beds	
	Rural/lodging tourism – without threshold	
SK	Establishments without threshold regarding the number of bed places	
	Statistical units (legal persons and natural persons/entrepreneurs with the Company Registration Number) – Accommodation establishments	

Data on the occupancy of tourist accommodation establishments are often published with more than a month delay. This is due to the allotted time the entities have to submit data on the occupancy of tourist accommodation establishments and the time needed to process data by services of official statistics.

In recent years, a growing demand for data on the rapidly changing situation in the tourism market has been observed. The use of non-statistical data, e.g. from web scraping, is an attempt to meet these expectations. The advantage of this data source is that complete data sets are available immediately after the end of the month which allows for rapid analysis. As a result of the work carried out, it was verified to what extent the combination of this data with official data on tourist accommodation establishments will affect the velocity and accuracy of estimates of the occupancy of tourist accommodation establishments.

Data sources

Due to the fact that the work on flash estimates is still ongoing, the following results have been calculated on data of Statistics Poland. The results of flash estimates of the occupancy of tourist accommodation establishments for other countries will be included in the final report.

Two sources of data were used to estimate the variables of interest (number of tourists accommodated and nights spent) from the scope of occupancy of tourist accommodation establishments:

- Web scraped data from accommodation booking portals.

  Data are scraped with daily frequency and include information on: location (address and/or coordinates), type of establishment, star rating, guest review rating and price of accommodation. Scraped data are used to calculate monthly statistics of continuous variables

such as price broken down by type of establishment and location. Web scraping data includes information on all establishments (but not necessarily information on the number of bed places or rooms offered).

- Data from a monthly statistical survey on the occupancy of tourist accommodation establishments.

  Data from the survey include information on accommodation establishments including number of tourists accommodated, the number of nights spent and for some types of establishments also the number of rented rooms. The survey covers establishments with 10 or more bed places - unlike in the data from the booking portals that covers all establishments It is, therefore, not possible to distinguish between small and large establishments.

Estimation procedure

The general estimation procedure can be described in the following steps:

- Preparation of time series of statistics from the survey of the occupancy of the tourist accommodation establishments.
- Clearing and formatting of data from web scraping.
- Aggregation of data from web scraping – statistics of price and number of offers by region and type of accommodation groups (Statistical Classification of Economic Activities in the European Community - NACE)
- Preparation of time series of statistics from web scraping at the same frequency as the time series from the survey
- Selection of a set of explanatory variables from the set of web scraping statistics based on substantive premises and statistical methods.
- Model construction and verification.
- Forecasting based on the model.

3.1 Case Study: Poland

The monthly results of the survey on the occupancy of tourist accommodation establishments are merged into one database to form time series. The resulting time series are the basis for flash estimates of the occupancy of tourist accommodation establishments. The second element are the daily web scraped data from accommodation booking portals. Daily web scraped data from accommodation booking portals are the second element. Daily data must be aggregated to form monthly statistics with variables such as the price or the number of offers broken down by, e.g. type of establishment and location. For continuous variables such as price, it is worth generating several different statistics, e.g. the first and the third quartile, median and mean.The flexibility of building a model is increased when more explanatory variables are available. In this way, monthly time series are also compiled.

Data from both sources are compiled by year and month. It should be noted that the data from web scraping can be obtained almost on an ongoing basis while the data from the survey are available with a delay. The missing values are the variables to forecast. Below is an example of how such a combined database may look like. The time series *Tourist* and *Nights spent* come from the survey of

accommodation establishments, while the remaining ones from the web scraping of accommodation portals (*hotels.com*).

Table 21. Example of a combined database of aggregated data

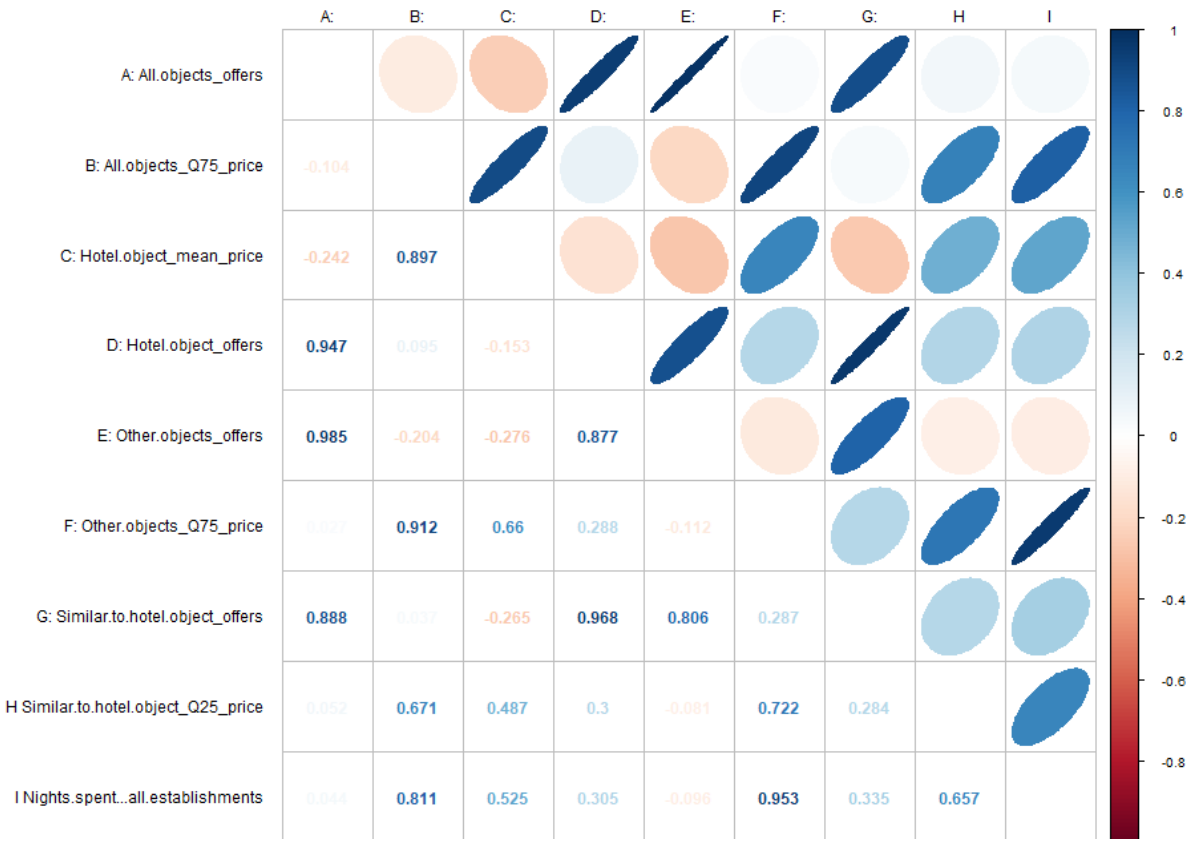
Year	Month	Mean price [EUR]	Offers	The first quartile of prices [EUR]	Median price [EUR]	The third quartile of prices [EUR]	Tourists	Nights spent
2019	1						79 127	218 394
2019	2	54.64	382	38.56	49.50	64.43	77 708	227 272
2019	3	54.33	1 217	37.81	49.50	62.24	81 887	217 349
2019	4	52.34	1 111	37.60	49.50	62.19	81 467	217 559
2019	5	57.27	1 008	42.29	51.49	64.43	120 846	309 035
2019	6	57.14	992	42.29	49.75	64.43	126 468	331 442
2019	7	61.05	620	42.29	52.37	68.98	148 712	488 201
2019	8	58.69	906	42.29	52.70	67.98	167 351	517 699
2019	9	58.16	988	37.81	51.49	69.40	-	-
2019	10	59.04	717	38.56	53.73	64.43	-	-

Cells marked by (-) indicate values for which estimates will be calculated.

A set of explanatory variables is determined for each explained variable. The developed algorithm determines these variables in two stages.

In the first stage, a correlation matrix is used. This allows only the variables best correlated with the explained variable to be selected as potential explanatory variables for the model.
Note that none of the variables from the scraped data relate directly to the use of the establishment. However, it can be assumed that in the time series obvious connections between the demand, supply and price of accommodation services will be observed. The higher the demand for accommodation services, the higher the average prices should be and the lower the number of offers. In fact, the following correlogram confirms these assumptions (see Figure 18).

Figure 18. Correlogram for variables from web scraping [A-H] and survey on tourist establishments [I]

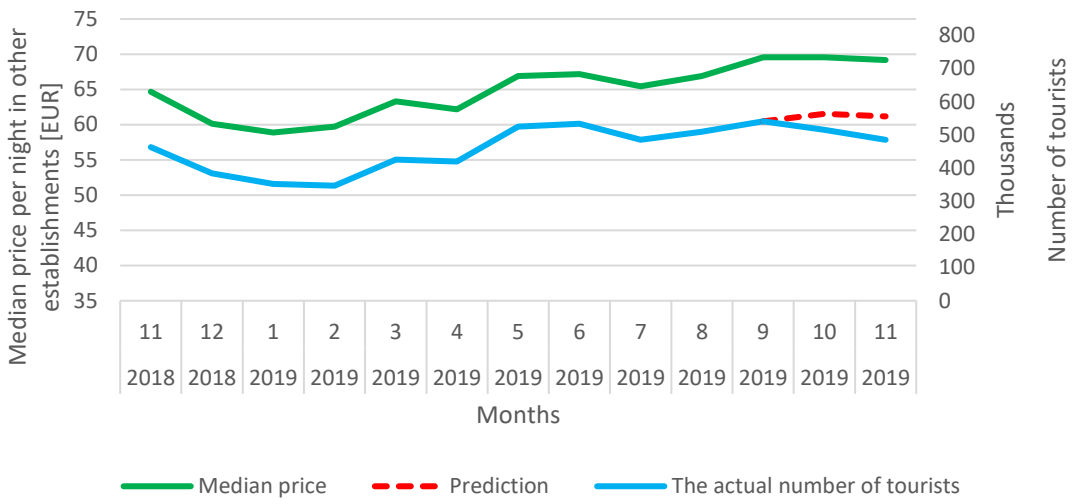


From the price statistics group, the one that is best correlated with the explained variable is selected. After such preliminary screening of variables with poor prognostic properties, the Hellwig method of the capacity of indicators' information is used during the second stage.

Then a model based on selected variables is built allowing to estimate the missing values. The strength of the proposed approach should be the selection of variables, not the model itself. Based on the preliminary results, it can be stated that even the least squares linear regression model has good prognostic properties.

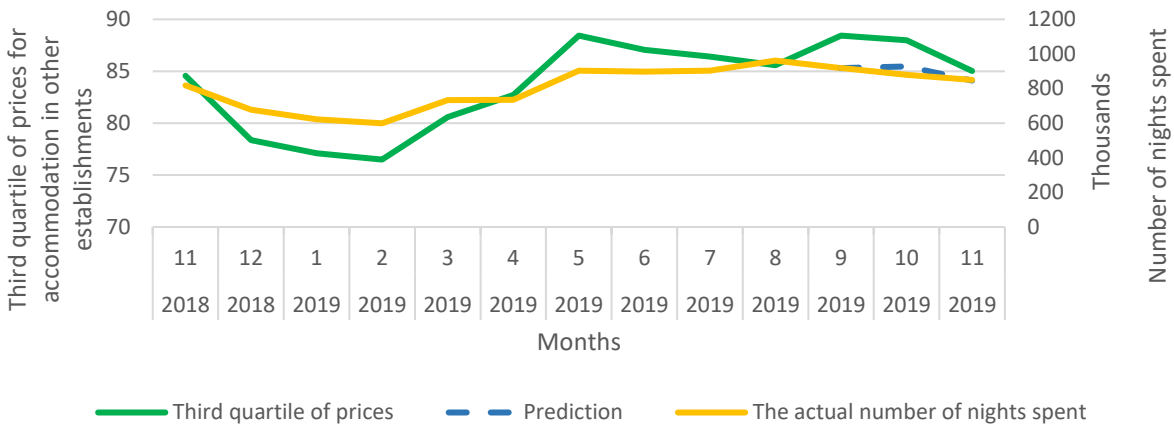
The Figure 19 presents the time series of the total number of tourists in the Mazowieckie region together with the estimation (the dotted line in red) and the independent variable (median price) of the model on the basis of which this forecast was calculated. The correlation between these variables in the analysed period amounted to 0.98. Therefore, the adjusted R-squared amounted to 0.96. The mean absolute percentage error (MAPE) of the estimated model amounted to 2.24%.

Figure 19. Time series of the number of tourists with a prediction based on the median price



The model to estimate the number of nights spent was characterized by adjusted R-squared of 0.899 and MAPE 3.09%. The correlation coefficient of the estimated dependent variable with the independent variable (the third quartile of prices) in the analysed period amounted to 0.95. Figure 20 presents time series of these variables in the analysed period.

Figure 20. Time series of nights spent together with prediction based on the third quartile of prices



In the future, when longer time series are available, it will be possible to use different types of models (e.g. ARIMAX models).

4. Conclusions

1. During the process of combining data obtained from web scraping with the survey frame of tourist accommodation establishments, the inconsistent address data have been the biggest obstacle. Therefore, using the method of combining data by geographical coordinates proved to be a better solution than the method based on address data.
2. On the basis of the implemented works, it has been shown that during the process of combining data it is necessary to use technology that allows geolocation of address data. For this purpose, it is reasonable to use the *HERE Maps API*, which has current maps for all European countries and extensive documentation on embedded functionalities.
3. The operation of the proprietary tool built on the basis of *HERE Maps API* allowed the generation of geographical coordinates in all countries involved in the WPJ despite the diverse structures of the survey frames. In many partner countries, this has led to the identification of new accommodation establishments that are not part of their survey on tourist accommodation establishments. The tool created as part of the WPJ project will be an important element of the built prototype.
4. Inventory of tourism related data sources has allowed for the development of a flow model for estimating trips and expenditure. This concept combined data from statistical surveys as well as big data and administrative sources using machine learning methods such as Regression Trees (RT), Support Vector Machine (SVM), Random Forest (RF), Predictive Mean Matching (PMM), Bayesian Linear Regression (BLR), Random Indicator for Nonignorable Data (RI). The methods allowed the identification of 65 new destination countries of trips of Poles that did not appear in the sample survey and to estimate expenditures related to trips to these countries.
5. Currently, in many countries of the European Union, data on the number of tourists and overnight stays are calculated on the basis of a survey on the use of tourist accommodation establishments with a long delay. New sources identified as a part of the work will allow the development of flash estimates using econometric models and obtaining preliminary results within T + 1.The use of models based on historical data (e.g. ARIMAX models) for flash estimates in the field of tourism requires long time series. It is recommended to use simple econometric models when long time series are not available. Preliminary results based on simple econometric models were characterized by small prediction errors. The calculated MAPE index (mean absolute percentage error) for the data collected during the project amounted to 3%.
6. Given that this is a pilot track project, the results obtained while trying to integrate multiple data sources are somehow exploratory. However, they lead to a clear improvement in timeliness, completeness and consistency in the monitoring of the supply and demand sides of the Tourism Accommodation Establishments Statistics.

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